Exact Wave-Function Solution and Spurious-Mode Discrimination for Completely Free Annular Sector Plates

Garrek McCune[a,*], Daniel Stutts[a]
[a] *Mechanical and Aerospace Engineering,*
Missouri University of Science and Technology, Rolla, MO 65409, USA
[*]Corresponding author. E-mail: ghmkfh@umsystem.edu

DRAFT — September 5, 2026

Abstract

The Seok–Tiersten wave-function boundary-determinant method is extended from the cantilevered annular sector to the completely free (FFFF) configuration, in both out-of-plane (flexural) and in-plane (extensional) motion. Frobenius-series wave functions satisfy the radial-edge conditions exactly; a two-edge variational functional enforces the circumferential conditions and decouples into even and odd parity blocks. Releasing both circumferential edges leaves them enforced only weakly, which admits spurious determinant zeros alongside the natural frequencies, so discriminating the two becomes the central problem of the extension. The two motion types reach different states of completeness: flexure gets a portable, two-sided residual screen that decides a candidate from its boundary residuals alone, on a cut calibrated once and unchanged since, while extension's pointwise residual is one-sided on its own. Two further extensional instruments close that gap: a portable mode-shape bar that still requires finite-element eigenvectors, and a second, FE-free criterion, an angle between the determinant's null direction and its own traction-minimizing direction, that certifies a root from the boundary determinant alone and is cluster-confirmed at four geometries and on a full blind adjudication set of 122 candidates (70 of 77 FE modes resolved blind, 73 once three nearest-frequency pairing defects the criterion itself flagged in advance are corrected against independent finite-element eigenvectors; zero duplicate zeros ever accepted). With the §5 instruments, the flexural tables agree with finite elements to 0.64% (isotropic) / 1.08% (orthotropic), with an independent literature recovery of 6 of 6 targets at a third geometry. The extensional tables list 26 solver roots covering 27 FE modes (one root serves the 1114.9/1115.3 Hz pair; worst unique-partner error 0.052%) and agree with two published semi-analytical solutions. A two-sided extensional bar on identity-weighted mode-shape correlation (threshold 0.744) classifies REAL-like versus ARTIFACT-like candidates at three $\nu = 0.30$ geometries and at all 24 extensional keys of a $\nu = 0.35$ sweep; at $2\Theta/\pi = 1.25$ and 1.50 every tight FE-frequency match is ARTIFACT-like. Two limits are reported with the rest: the tabulated frequencies are not a certified single-pass product, since a fully-default detector misses 4 of 26 in-plane modes, later recovered as genuine roots of the same determinant; and one flexural FE target at $\Omega_{\mathrm{lit}} = 111.46$ remains an open assignment.

Keywords: annular sector plate; free vibration; exact solution; free boundary; in-plane vibration; spurious modes; discrimination instruments; modal assurance criterion.

1 Introduction

Annular and sector-shaped plates recur as vibrating components wherever a full disk is unavailable or unconstrained: turbine and compressor blade-disk sectors, brake and clutch friction elements, disk-type resonators, and gear or bearing-retainer segments. The completely free (FFFF) boundary condition treated here is the natural idealization for such a component tested, or operating, without an external mount. Exact solutions provide a reference standard in structural vibration analysis that approximate methods cannot generate. General-purpose global-basis techniques dominate engineering practice, but their error is bounded only empirically, against a finite-element mesh whose own convergence is itself asymptotic. An exact solution instead supplies a reference frequency whose error is set by the retained circumferential basis and the working precision, not by a spatial mesh, giving approximate methods something to check their convergence and completeness against. That role matters most where approximate bases are weakest: dense, near-degenerate spectral regions, such as the in-plane spectrum of a sector plate, where basis truncation and mesh refinement both struggle to resolve closely spaced modes with confidence.

The exact-wave-function method of Seok, Tiersten, and (for the rectangular geometry) Scarton [1, 2, 3, 4] is one such reference solution for annular sector and rectangular plates: the radial (or axial) dependence is carried by exact wave functions rather than a truncated basis, the circumferential edge conditions are enforced through a variational functional, and natural frequencies emerge as the zeros of a finite characteristic determinant. The method is exact in the direction it treats analytically and spectrally convergent in the finite direction, and the 2004 papers validated it against contemporary numerical benchmarks to good agreement. It stopped, however, at a single boundary configuration, the clamped-free (cantilever) sector, and its numerical machinery (dispersion-root tracking, branch selection, determinant assembly) was never released, leaving the method effectively unusable by anyone outside the original derivation.

The fully free (FFFF) sector is, among the classical boundary conditions, the hardest one to extend the method to. The reason is structural: a clamped edge imposes kinematic (essential) constraints that the assumed solution can be built to satisfy exactly, leaving only a natural (weak) condition at the opposite, free edge; the cantilever's single free edge is accordingly the method's easiest case. Releasing both circumferential edges removes every kinematic constraint from the circumferential direction: both edges' conditions become natural, enforced only in the finite Galerkin sense that the retained wave-function basis allows, and the configuration additionally admits rigid-body modes at zero frequency. Weak enforcement on both sides at once admits spurious determinant zeros (nearly displacement-free evanescent combinations that satisfy the finite projection without satisfying the physical boundary conditions pointwise), a failure mode with no counterpart in the cantilever case: the clamped edge's exact kinematic satisfaction leaves no room for it to arise there. Discriminating these spurious zeros from genuine modes is therefore the central problem the FFFF extension must solve; the instruments developed in §5 address it directly. A companion paper [18] treats the rectangular FFFF problem, which needs a new assembler (closed-form axial waves, four free corners) rather than a change of this paper's boundary list.

This paper (i) documents an arbitrary-precision reimplementation, to be released as open source alongside this paper, that reproduces the published cantilever tables from scratch; (ii) derives and validates the extension of the boundary determinant to two free circumferential edges, including the edge-orientation bookkeeping this requires; (iii) identifies the resulting spurious-zero problem and develops a pointwise edge-residual screen that is robust and calibrated for flexural motion and is only a one-sided, calibrated-adjudication instrument for extensional motion, together with a basis-enlargement direction rule, an unconstrained fine-scan arbitration step, and a two-sided extensional bar on identity-weighted mode-shape correlation for use when FE eigenvectors are available; (iv) reports FFFF mode tables obtained with those instruments (not a certified single-pass detector) FE-checked in flexure under a portable residual screen, FE-checked in extension under a one-sided, geometry-specific residual screen, and independently checked in extension against two published semi-analytical solutions by direct singularity-depth agreement, with the mode-shape bar applied as a separate instrument; (v) shows that every REAL-like flexural candidate has its own independent finite-element partner at three further geometries, together with a coarser 24-geometry nearest-neighbor sweep run for both motion types; and (vi) introduces a second, FE-free extensional certification criterion (SUBDOM) that decides a candidate from the boundary determinant alone.

2 Related work

The contemporary sector-plate vibration literature is dominated by general-boundary-condition frameworks built on approximate global bases. Modified Fourier–Ritz formulations with supplementary edge functions, combined with the Ritz procedure, underpin unified first-order-shear-deformation solutions for functionally graded [5] and laminated composite [6] circular, annular, and sector plates, and benchmark tables for graded sandwich sector plates on two-parameter elastic foundations [7]. For in-plane (extensional) motion specifically, the less-studied problem and the one in which basis truncation is hardest to detect because modes cluster densely, recent frameworks include isogeometric analysis with NURBS bases [8] (its own isotropic free-free table is matched directly in §6.9 below, with no FE model standing between the two), a generalized Fourier series treatment of annular plates with arbitrary spring-modeled edge restraints [9], a modified Fourier–Ritz solution for orthotropic circular, annular, and sector plates under general boundary conditions [10] (its own published in-plane orthotropic free-free table is checked the same way, directly, in §6.8 below), and a semi-analytical variational/multi-segment method with Jacobi-polynomial expansions and penalty-enforced elastic constraints for composite laminated sector and annular plates [11]. These methods owe their popularity to flexibility: arbitrary classical or elastic edge conditions are handled by one formulation, and convergence is demonstrated empirically against finite elements or against each other.

This body of work has no exact counterpart (§1): reference data come almost exclusively from finite-element models, and the in-plane spectrum's dense clusters and near-degenerate pairs are precisely where that is least trustworthy. The exact-wave-function line of Seok and Tiersten [1, 2, 3, 4] can supply that counterpart, but it stopped in 2004 at the cantilever configuration. Independent benchmark results exist for particular free configurations, notably the orthotropic annular free-free material properties of Shi, Liang, Wang and Teng [14], used as the basis for an independently-run finite-element benchmark here (§6.2), Wang, Shi, Liang and Ahad's own published in-plane orthotropic free-free table [10], again matched directly (§6.8), and Shi, Shi, Li, Wang and Han's isotropic flexural free-free frequencies for annular sector plates [15], matched directly in §6.10; but no systematic exact framework spans general boundary conditions.

A separate body of work speaks directly to the difficulty this extension runs into. Eigenformulations that impose their boundary conditions through a projected boundary operator rather than pointwise are known to return spurious eigenvalues alongside the true ones, and the mechanism Chen, Lee, Chen and Kuo [13] give for it is the one at work here: when part of the constraint is dropped, the admissible solution space grows, and the characteristic determinant acquires zeros that no physical mode corresponds to. Lee and Chen [12] analyze this for the free vibration of circular plates under a real-part boundary integral equation method, where discarding the imaginary-part kernel halves the computation at the cost of spurious roots, and filter them out by adjoining constraint equations at exterior collocating points; Ref. [13] treats annular and confocal elliptical membranes by dual BEM and the null-field BIEM, separating the two populations through a singular value decomposition of the discrete system and showing that true eigenvalues depend on the boundary condition while spurious ones depend on the formulation. The present setting differs in method — a Frobenius wave-function determinant rather than a boundary integral equation, with the incompleteness sitting in the circumferential Galerkin projection instead of a real-part kernel — but the diagnosis transfers, and so does the shape of the remedy: the instruments of §5 are a pointwise-residual and mode-shape analogue of the filtering those papers apply in their own setting.

3 Problem statement and mathematical model

3.1 Geometry and governing equations

Consider an annular sector plate of inner radius R_i, outer radius R_o, thickness H, sector angle 2Θ, density ρ, Young's modulus E and Poisson ratio ν (the orthotropic generalization is summarized in §3.4). For time-harmonic motion at circular frequency ω, the out-of-plane displacement $w(r, \theta)$ of the classical thin-plate model satisfies

$$D\,\nabla^4 w \;=\; \rho H \omega^2 w, \qquad D = \frac{EH^3}{12(1-\nu^2)}, \tag{1}$$

and the in-plane displacements (u_r, u_θ) satisfy the plane-stress Navier equations

$$\frac{E}{1-\nu^2}\Big[\nabla(\nabla\cdot\mathbf{u}) - \tfrac{1-\nu}{2}\,\nabla\times(\nabla\times\mathbf{u})\Big] \;=\; -\rho\,\omega^2\mathbf{u}. \tag{2}$$

Equations (1)–(2) correspond to Ref. [1] Eq. (10) (out-of-plane, obtained there from the moment/shear-resultant divergence after imposing zero transverse shearing strain on Mindlin's power-series plate hierarchy, which returns classical Kirchhoff theory rather than a first-order shear-deformable one) and Ref. [2] Eqs. (9)–(10) (in-plane, from the stress-resultant divergence). Both papers nondimensionalize the radius as $\bar{r} = \pi r/2b$, $\bar{r}_0 = \pi r_0/(2b)$, $r^* = \bar{r} - \bar{r}_0$ ($b = (R_o - R_i)/2$, $r_0 = (R_o + R_i)/2$) and the frequency as $\bar{\Omega} = \omega/\bar{\omega}$ with $\bar{\omega} = (\pi/2b)\sqrt{c^*_{66}/\rho}$ (Ref. [1] Eqs. (26)–(28); Ref. [2] Eqs. (23)–(25) use the same $\bar{r}$, r^* and an analogous $\bar{\Omega}$). Ref. [1] Eq. (27) additionally defines the thickness-dependent scale $\bar{k}$, and $\bar{\Omega}$ enters the out-of-plane dimensionless equation (Ref. [1] Eq. (29)) only through the product $\bar{k}\bar{\Omega}$; that product, not $\bar{\Omega}$ itself, is accordingly what Ref. [1]'s own Table 2 tabulates and what the present work's out-of-plane tables report as their raw column. The in-plane problem has no such factor and reports $\bar{\Omega}$ directly. Both are written Ω below where no ambiguity arises. The out-of-plane governing equation, after separating $w = \bar{W}(\bar{r})\sin(\xi\theta)e^{i\Omega\bar{\omega}t}$, becomes the fourth-order ODE Ref. [1] Eq. (25); the in-plane displacements u_r, u_θ separate as $\{F(\bar{r})\sin(\zeta\theta),\, G(\bar{r})\cos(\zeta\theta)\}e^{i\Omega\bar{\omega}t}$ (Ref. [2] Eqs. (19)–(20)), giving the coupled second-order system Ref. [2] Eqs. (21)–(22). The material constants entering both are $\bar{D} = 2h^3c^*_{11}/3$, $\bar{\nu} = c^*_{12}/c^*_{11}$, $T = (2c_{66}/c^*_{11}) + \bar{\nu}$, $R^* = c^*_{22}/c^*_{11}$ (Ref. [1] Eq. (16)), reducing to $T = R^* = 1$, $\bar{D} = D$, $\bar{\nu} = \nu$ for the transversely isotropic case used throughout the present tables.

The radial edges $r = R_i, R_o$ carry the papers' exactly satisfied conditions; the circumferential edges $\theta = \pm\Theta$ carry, in the present work, the free conditions: vanishing bending moment and effective (Kelvin–Kirchhoff) shear for out-of-plane motion, and vanishing normal and shear tractions $T_{\theta\theta} = T_{\theta r} = 0$ for in-plane motion. The isotropic free-free tables of §6 use a single physical plate ($R_i = 4$ m, $R_o = 8$ m, $H = 0.08$ m, $2\Theta = \pi/2$, steel $E = 210$ GPa, $\nu = 0.30$, $\rho = 7800$ kg/m^3), for which $\bar{\omega} = 2527.351$ rad/s and $\bar{k} = 32.617$: in-plane raw output is already $\bar{\Omega} = \omega/\bar{\omega}$, out-of-plane raw output is the product $\bar{k}\bar{\Omega}$. The conversion algebra and the eight-row numerical cross-check against Table 6 are in Supplementary Material, §S.3.6.

Table 1: Nomenclature: barred and nondimensional quantities introduced in this section (isotropic case $T = R^* = 1$, $\bar{\nu} = \nu$, $\bar{D} = D$ unless noted), followed by the detector and discrimination-instrument symbols used from §4 onward.

Symbol	Definition
$b = (R_o - R_i)/2$	half radial width
$r_0 = (R_o + R_i)/2$	center (expansion) radius
$\bar{r} = \pi r/2b$	nondimensional radius
$\bar{r}_0 = \pi r_0/2b$	nondimensional center radius
$r^* = \bar{r} - \bar{r}_0$	shifted radial expansion variable
$\bar{\Omega} = \omega/\bar{\omega}$	nondimensional frequency of Refs. [1, 2]
Ω	raw solver frequency: $\Omega = \bar{\Omega}$ (in-plane), $\Omega = \bar{k}\bar{\Omega}$ (out-of-plane); Supplementary Material, §S.3.6
$\bar{\omega} = (\pi/2b)\sqrt{c_{66}^*/\rho}$	in-plane frequency scale
$\bar{k}$	out-of-plane (thickness-dependent) wavenumber scale (Supplementary Material, §S.3.6)
$\Omega_{\mathrm{lit}} = \omega R_o^2\sqrt{\rho H/D}$	classical out-of-plane literature frequency ratio
$\Omega_{\mathrm{Wang}},\ \Omega_{\mathrm{Qin}}$	source-specific in-plane frequency parameters, each defined where first used (§6.8, §6.9)
$\bar{D} = 2h^3 c_{11}^*/3$	nondimensional flexural rigidity ($h = H/2$)
$\bar{\nu} = c_{12}^*/c_{11}^*$	nondimensional Poisson-type ratio
$T = (2c_{66}/c_{11}^*) + \bar{\nu}$	dimensionless orthotropic constant
$R^* = c_{22}^*/c_{11}^*$	dimensionless orthotropic constant
$\sigma_{\min}$	smallest singular value of $\mathbf{K}(\Omega)$; "depth" means $\log_{10}\sigma_{\min}$ (§4)
n_{dofs}	generalized coordinates retained in $\mathbf{K}$ (§4)
Ω^*	a converged, bracket-stable root of $\det\mathbf{K}(\Omega) = 0$
$r_M,\ r_V$	flexural moment / effective-shear edge-residual ratios (§5)
$r_{\theta\theta},\ r_{\theta r}$	extensional normal- / shear-traction edge-residual ratios (§5)
MAC	identity-weighted modal assurance criterion (§6.6)
$\mathbf{D}, \mathbf{T}$	displacement- / traction-weight matrices of $\mathbf{K} = \mathbf{D}^{\mathsf{T}}\mathbf{T}$ (§6.7); bold, distinct from the scalars D, T above
$\eta(c),\ \eta(\Omega)$	traction-to-displacement ratio at a coefficient vector, and its minimum over all of them (§6.7)
$\gamma,\ p$	degeneracy-pooling factor ($\gamma = 4$) and the resulting number of pooled lowest-traction directions (§6.7)
SUBDOM	interior-field subspace-dominance criterion; the call is SUBDOM > 0.70 (§6.7)

3.2 Exact radial wave functions

Separating $w = W(r)\,\{\sin, \cos\}(\xi\theta)$ (and correspondingly for u_r, u_θ with transverse wavenumber ζ) reduces (1)–(2) to ordinary differential equations in r whose solutions are constructed as Frobenius series about the center radius r_0. For each frequency the admissible transverse wavenumbers are the roots of the radial-edge dispersion relation; the root set comprises real, imaginary, and complex families, all of which must be retained. The basis is assembled by a census-and-fill procedure that tracks every branch across frequency; dropping a family produces silent mode loss, and branch completeness proved to be the practically difficult part of replication. Following Ref. [1] §3 (out-of-plane) and Ref. [2] §3 (in-plane), each expansion is taken about r_0, not the singular point $r = 0$, guaranteeing four independent solutions regular on the plate; the recursion for the series coefficients, which the source papers describe as "much too cumbersome to include" in closed form, is carried numerically to convergence at working precision (`mpmath`, 40 digits) instead of being truncated at a fixed low order as the papers' fixed-precision Maple calculation could tolerate. The full indicial-equation and four-branch selection detail, and the branch-census construction (tracking what the papers' own dispersion curves, Ref. [1] Fig. 2, exhibit), are given in Supplementary Material, §S.1.

3.3 Two-edge variational functional and parity decomposition

The circumferential-edge conditions are enforced through the papers' variational functional, projected onto the retained basis of n wave functions, giving the boundary matrix $\mathbf{K}(\Omega) \in \mathbb{R}^{n\times n}$;

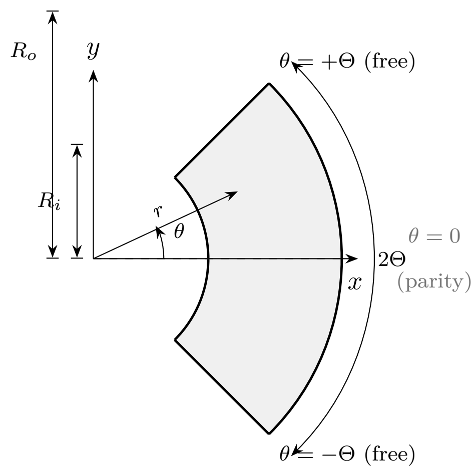


Figure 1: Annular sector: radial edges $r = R_i, R_o$ exact; circumferential edges $\theta = \pm\Theta$ free. Primary benchmark $2\Theta = \pi/2$; §6.5 also treats $2\Theta = \pi$ and $R_i/R_o = 0.6$; §6.4's sweep spans further radius ratios and sector angles on the same schematic.

natural frequencies satisfy $\det \mathbf{K}(\Omega) = 0$. For the cantilever, the clamped edge contributes kinematic rows and the free edge natural rows. For the free-free plate both edges contribute natural rows, and the two contributions carry opposite orientation signs; with the orientation carried correctly the matrix decouples exactly into even and odd angular-parity blocks, $\mathbf{K} = \mathrm{diag}(\mathbf{K}^e, \mathbf{K}^o)$, halving the computation and assigning each mode a definite parity. (If the sign is instead dropped, the same-parity contributions cancel identically for arbitrary radial content: a structural cancellation demonstrated algebraically later in this subsection.) The papers derive this functional for the single-free-edge cantilever, in which one circumferential edge ($\theta = \Theta$) is free and the other ($\theta = -\Theta$) is clamped: once the radial-edge conditions, already satisfied exactly by the wave functions of §3.2, are used to eliminate the bulk term, what remains is the corner-bearing form Ref. [1] Eq. (43) for out-of-plane motion (Ref. [2] Eq. (48) for in-plane), giving the assembled system $\mathbf{KX} = 0$ (Ref. [1] Eq. (46)) with $\mathbf{X}$ the per-branch, per-parity amplitude coefficients (Eq. (47)). Eq. (43)'s tail carries *two* corner terms, and it is essential to the free-free derivation to see that they are not the same term at two locations: each is a jump contribution over the two radial edges, $r^* = \pm\pi/2$, but evaluated at a single, different circumferential edge apiece: coefficient $+(T - \bar{\nu})$ at the *clamped* edge $\theta = -\Theta$, and coefficient $-2(T - \bar{\nu})$ at the *free* edge $\theta = \Theta$ (transcribed verbatim in Supplementary Material, §S.2, Eq. (B.1)). The clamped-edge term arises from the paper's Lagrange-type substitution enforcing $w = \partial w/\partial\theta = 0$ at $\theta = -\Theta$; the free-edge term is the genuine free-edge natural boundary contribution. These are two structurally different derivations that happen to multiply the same combination $(T - \bar{\nu})$. The same wall/tip asymmetry recurs in three source-paper cases: the present annular Eq. (43), the rectangular out-of-plane case (Ref. [3] Eq. (53)), and the rectangular in-plane case (Ref. [4] Eq. (36)). The free-free plate's $\theta = -\Theta$ corner term must therefore reuse the *free*-edge formula, $-2(T - \bar{\nu})$, re-evaluated at $\theta = -\Theta$ with the correct relative orientation, not Eq. (43)'s clamped-edge coefficient. The remainder of this subsection states the resulting free-free corner-term formula and proves the same-parity cancellation of the mis-oriented assembly. The closed-contour (n, s) table, the four corner jumps, and the surface-term substitution this builds on are in Supplementary Material, §S.2.

The corner-jump derivation. Ref. [1] Eq. (6), reproduced in full in Supplementary Material, §S.2 (Eq. (B.2)), has its own asymmetric corner factors (-1 at $\theta = -\Theta$, $+2$ at $\theta = \Theta$), which reflect that, in the cantilever's contour C_N, a free $\theta=\Theta$ corner is where two free edges meet, while the $\theta=-\Theta$ corner sits where C_N terminates at the wall. For the free-free case C_N becomes the full closed boundary, and the corner coefficient follows from Eq. (1a)'s fourth term $\int_{c_N} ds\,(n_a\tau^{(1)}_{a\beta}s_\beta\,\delta w)_{,s}$: n_a, s_b are discontinuous at each corner even though the physical stress $\tau^{(1)}_{r\theta}$ is continuous there. Writing $F\equiv n_a\tau^{(1)}_{a\beta}s_\beta$ on each segment of that closed contour (Supplementary Material, §S.2) and taking the jump as incoming F minus outgoing F gives magnitude 2 at every corner, alternating in sign between the two edge pairs: the $\theta=\Theta$ pair reproduces Ref. [1]'s printed cantilever corner term exactly, and the $\theta=-\Theta$ pair, now smooth-turn corners of the closed contour, gives the free-edge coefficient -2 with the same orientation sign already used for every corner, so the identical ±1 orientation factor that fixes the sign of Eq. (B.3′) (Supplementary Material, §S.2) also fixes the corner term's. Substituting the remaining constitutive relation this term needs,

$$\tau^{(1)}_{r\theta}=-\bar{D}(T-\bar{\nu})\frac{\partial^2}{\partial r\partial\theta}\left(\frac{w}{r}\right), \qquad \text{(Ref. [1] Eq. (15))}$$

and the same nondimensionalization as Supplementary Material, §S.2 turns this jump result into the free-free analogue of Eq. (43),

$$+2(T-\bar{\nu})\left[\frac{\partial^2}{\partial r^*\partial\theta}\left\{\frac{w}{r^*+\bar{r}_0}\right\}\right]^{r^*=\pi/2,\,\theta=-\Theta}_{r^*=-\pi/2,\,\theta=-\Theta}-2(T-\bar{\nu})\left[\frac{\partial^2}{\partial r^*\partial\theta}\left\{\frac{w}{r^*+\bar{r}_0}\right\}\right]^{r^*=\pi/2,\,\theta=\Theta}_{r^*=-\pi/2,\,\theta=\Theta}=0. \qquad (3)$$

Magnitude and alternation follow from the contour argument just given and from an independent Green's-identity argument (a bilinear-form check on the classical constant pure-twist field, Supplementary Material, §S.2) that reproduces the same free-edge magnitude and factor of 2 by a code- and FE-independent route. What neither argument assigns on its own is which isolated edge carries the "+" label — but that is a convention of the overall stationarity condition rather than physical content: multiplying the whole equation by -1 swaps both labels and moves no frequency. What has to be consistent is the *relative* sign between the corner term and the surface term inside the same equation, and it is: the identical ±1 orientation factor fixes both, and the implemented assembly applies that one rule on both free edges. No finite-element result or published value enters at this step; §6's FE comparison tests the completed assembly rather than supplying any coefficient in it.

Orientation and the same-parity cancellation proof. Write each retained basis function's angular factor as $\sin(\xi\theta+q\pi/2)$, $q\in\{0,1\}$ ($q=0$: the sin, angularly ODD family; $q=1$: the cos, angularly EVEN family, since $\sin(\xi\theta+\pi/2)=\cos(\xi\theta)$). Evaluating the identical factor at the mirror edge $\theta\to-\theta$,

$$\sin(-\xi\theta+q\pi/2)=(-1)^{1-q}\sin(\xi\theta+q\pi/2) \quad\Longleftrightarrow\quad \begin{cases} q=0: & \sin(-\xi\theta)=-\sin(\xi\theta)\ \ \text{(sign flips)},\\ q=1: & \cos(-\xi\theta)=\cos(\xi\theta)\ \ \text{(sign fixed)}.\end{cases}$$

A boundary-matrix entry K_{ij} pairs a test function i and a trial function j, each carrying its own parity label $q_i, q_j\in\{0,1\}$, and the free-edge pairing mixes a stress factor of one with a displacement factor of the other. Writing $\sigma(q)=-1$ for $q=0$ and $+1$ for $q=1$, the free-edge pairing is a mixed sin/cos projection (one function's stress factor times the other's displacement factor), so it is odd in the edge sign for SAME-parity pairs. The unsigned functional evaluated at $\theta=-\Theta$ therefore equals $-\sigma(q_i)\sigma(q_j)$ times its value at $\theta=\Theta$: -1 whenever $q_i=q_j$ (SAME parity) and $+1$ whenever $q_i\neq q_j$ (CROSS parity), independent of which specific parities are involved. The correctly oriented free-free assembly combines the two edges' contributions with a relative orientation sign $+1$ at

$\theta = \Theta$ and -1 at $\theta = -\Theta$ (the discrete trace of the functional's $[\,\cdot\,]_{-\Theta}^{\Theta}$ derivation, and the rule the deployed solver uses); the mis-oriented assembly sums both edges with the same sign instead. Carrying the $-\sigma(q_i)\sigma(q_j)$ relation through both cases:

$$\text{oriented: } K_{ij} \;\propto\; 1 + \sigma(q_i)\sigma(q_j) \;\Rightarrow\; \begin{cases} 2 \times (\text{raw}), & q_i = q_j \\ 0, & q_i \neq q_j, \end{cases}$$

$$\text{un-oriented: } K_{ij} \;\propto\; 1 - \sigma(q_i)\sigma(q_j) \;\Rightarrow\; \begin{cases} 0, & q_i = q_j \\ 2 \times (\text{raw}), & q_i \neq q_j. \end{cases}$$

The un-oriented assembly therefore makes every SAME-parity entry vanish identically and every CROSS-parity entry double, for arbitrary radial content; the free-free $\mathbf{K}$ becomes exactly anti-block-diagonal, $\mathbf{K} = \left(\begin{smallmatrix} 0 & A_{oe} \\ A_{eo} & 0 \end{smallmatrix}\right)$, rather than the correctly oriented, physically meaningful $\mathbf{K} = \mathrm{diag}(\mathbf{K}^e, \mathbf{K}^o)$.

In the mis-oriented matrix that cancellation is visible directly: $q = 0$ (sin-family) cancellations are bit-exact and $q = 1$ (cos-family) cancellations hold to working precision (∼10–18 digits at 40-digit arithmetic) because of the inexact binary $\pi/2$ phase. Mixed $\partial^2/\partial r^*\partial\theta$ cross-terms were checked against the assembled matrix.

For the in-plane problem the papers' double-root criterion carries over in a specific, non-trivial sense that is worth stating precisely rather than only qualitatively: Ref. [2] shows (§4) that the *unconstrained* functional, Eq. (48), already decouples into two distinct linear systems that both vanish at the same true eigenfrequency (a double root whose two branches carry non-unique amplitude ratios), an artifact of using the same functional form for solution and variation that also occurs for the cantilever beam (Ref. [2] footnote 3). The papers remove the non-uniqueness, not the double root's redundancy, by adjoining a Lagrange-multiplier mean-displacement constraint at the fixed edge (Eq. (52)) to obtain the augmented, non-doubly-rooted system $\hat{\mathbf{K}}\hat{\mathbf{X}} = 0$ (Eq. (54), size $2P+1$ including the multiplier); roughly half of $\hat{\mathbf{K}}$'s roots coincide with the unconstrained double roots (genuine) and the rest are constraint-induced spurious roots to be discarded. This is the source of the "necessary but not sufficient" status invoked in §5: exact constrained/unconstrained coincidence is the paper's own criterion for a genuine root, demonstrated in §5 to admit false positives once weak circumferential enforcement is also present.

3.4 Orthotropic constitutive extension

The polar-orthotropic reduction (Ref. [1] Eq. (16)) enters the solver only through the three dimensionless ratios $T = (2c_{66}/c_{11}^*) + \bar{\nu}$, $R^* = c_{22}^*/c_{11}^*$, $\bar{\nu} = c_{12}^*/c_{11}^*$; setting $T = R^* = 1$ recovers the transversely isotropic case used for all non-orthotropic tables in §6. From engineering constants $(E_r, E_\theta, G_{r\theta}, \mu_r)$ the polar-orthotropic reduction is

$$\mu_\theta = \mu_r \frac{E_\theta}{E_r}, \qquad R^* = \frac{E_\theta}{E_r}, \qquad T = \mu_\theta + 2G_{r\theta}\frac{1 - \mu_r\mu_\theta}{E_r}.$$

The orthotropic set used in §6.2 ($E_r = 40$ GPa, $E_\theta = 70$ GPa, $G_{r\theta} = 3.51$ GPa, $\mu_r = 0.3$) gives $T = 0.672859$, $R^* = 1.75$, $\mu_\theta = 0.525$. It is adapted from, and is not identical to, the set of Shi, Liang, Wang and Teng [14], who specify $E_r = 40E_\theta$ with the same E_θ, $G_{r\theta}$ and μ_r; §6.2 records why the benchmark is run at the adapted value. The solver's parity decomposition and root-discrimination instruments (§5) do not depend on isotropy.

4 Numerical method

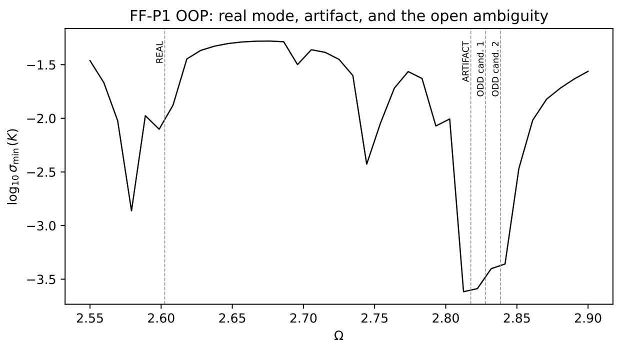
All series evaluation, wavenumber refinement, and linear algebra run in 40-digit arbitrary-precision arithmetic (`mpmath`); boundary-matrix entries span hundreds of orders of magnitude and fixed precision is not viable. Singularity is detected by the smallest singular value $\sigma_{\min}(\mathbf{K})$, not the determinant. The frequency axis is covered by a two-stage scan (coarse sweep with local zoom around prominence-selected candidates), followed by automatic re-scanning of suspiciously wide gaps at ten-fold resolution (a completeness device that recovered five genuine modes in the final tables), and each accepted dip is converged by golden-section refinement, chosen because $\sigma_{\min}$ is not smooth enough near a root to trust a gradient (it is the smallest singular value of a matrix assembled from Frobenius series and root-found transverse wavenumbers, not a closed-form function of Ω) and golden-section search only requires function evaluations and bracketing, not derivatives. Scans parallelize over frequency points using a standard multiprocess architecture, with each worker rebuilding its solver from scalar geometry/material constants instead of sharing solver objects across the process boundary; a complete blind first pass of one motion class costs a few hundred CPU-hours at 40 digits. The detection phase uses no prior knowledge of any kind: no seeded windows, no reference values, no preprogrammed regions of interest. A blind run at the production detector default (a 28-function basis, whole-window, no seeded regions; the discrimination instruments of §5 are separately calibrated at their own n=20 screening basis, and the two should not be confused) missed 4 of the 26 tabulated extensional modes. Targeted fine-scans recovered all four as genuine roots of the $n = 28$ determinant. Single-pass blind completeness is not certified at the production defaults; the §5 instruments are required for the method as published. A subsequent opt-in local split/anchor enhancement recovered those four unaided in the same configuration, but failed a required second-geometry blind pass and is not the package default (Supplementary Material, §S.3.1 and §S.4). The convergence of the frequency detector with basis size was checked for all 34 tabulated free-free modes (8 out-of-plane, 26 in-plane) by re-polishing each mode's own reference value at $n = 20$, 28, and 36 generalized coordinates. Twenty-four of the thirty-four labels change by less than 0.1% across all three basis sizes, and all but one (IP-06) by less than 0.25%. The remaining outliers trace to two distinct, understood mechanisms, not basis-truncation loss or non-convergence: a wide-window lock-on diagnostic artifact (the same mechanism discussed in §5 for fine-scan arbitration, recurring one layer up here), recovered in every case to within 0.5% of the seed by a tight-bracket re-polish; and, for several in-plane labels, the branch-selection routine correctly declining to complete the labeled basis size rather than admitting a branch that would corrupt the assembled matrix (confirmed directly: forcing completion anyway degrades $\sigma_{\min}$ from a genuine minimum to a shallow lock-on in every one of 32 test evaluations). With these mechanisms accounted for, all 34 modes change by less than 0.5% between $n = 28$ and $n = 36$, and the great majority by less than 0.1%; none of this calls the tabulated frequencies themselves into question, since they come from the production detector pipeline and are independently confirmed against finite-element benchmarks in §6. Full per-mode convergence tables and the basis-shortfall mechanism are in Supplementary Material, §S.3.1.

5 Spurious zeros and their discrimination

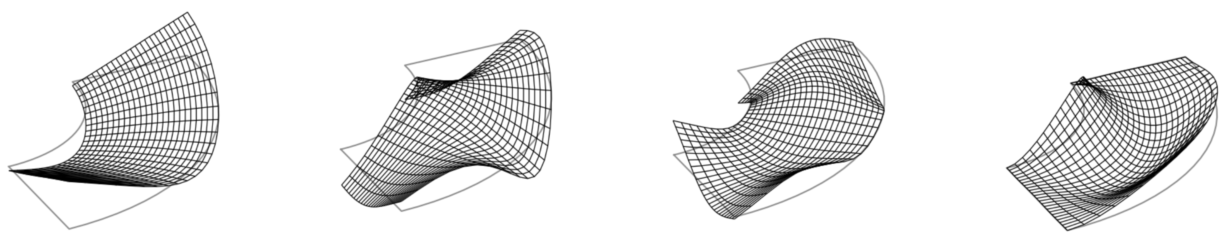
Free-free per-parity blocks enforce the free-edge conditions of a single edge on half the basis, and the determinant consequently acquires zeros that are not natural frequencies: nearly displacement-free evanescent combinations that annihilate the finite set of Galerkin projections while violating the boundary conditions pointwise. The phenomenon is not peculiar to this formulation; it is the wave-function-determinant counterpart of the spurious eigenvalues long documented for boundary-integral eigenproblems (§2), and, as there, mode shape rather than frequency is what ultimately separates the two populations. The residual screen is robust and calibrated for flexural motion; for extensional motion it is a one-sided, calibrated-adjudication instrument only. Three instruments are used. For flexural motion the screen separates physical and spurious zeros with no overlap at every basis size tested here, and by more than three orders of magnitude at the production basis; it is the primary discriminant. For extensional motion it is applied only at the calibrated production adjudication points, with marginal cases routed to the other two instruments (the portability characterization is detailed later in this section). Extensional results in the 24-geometry and second-geometry sweeps are therefore reported as raw residual ratios, not as verdicts. No extensional table in this paper is screened by a geometry-general residual threshold. (i) *Pointwise edge-residual screen*: the null vector of the accepted zero is reconstructed and the moment/shear (or traction) residuals are evaluated on a dense grid along the free edge, normalized by the peak displacement. Write r_M and r_V for the flexural moment and effective-shear residual ratios so defined, and $r_{\theta\theta}$, $r_{\theta r}$ for their extensional normal- and shear-traction counterparts: a ratio near zero means the free-edge condition is met pointwise, a ratio of order unity or larger means it is not. The labels "REAL-like" and "ARTIFACT-like" used throughout are the verdict these instruments return on a candidate, not a proof that it is or is not a natural frequency. For the flexural tables, at the production screening basis (n=20), the eight physical modes exhibit residual ratios r_M spanning 0.0025–0.037 and the six catalogued spurious zeros 74.9–348, a separation of more than three orders of magnitude. For extensional motion, at the same production adjudication points only, the corresponding ratios sit below $\sim$0.3 and above $\sim$4; that split is not used as a portable two-sided bar. (ii) *Basis-enlargement direction rule*: under an enlarged basis a physical root's pointwise residual improves while a spurious one degrades; this adjudicates marginal screen values. (iii) *Unconstrained fine-scan arbitration*: where two zeros lie within $\sim 10^{-3}$ in Ω, local refinement inherits the blind spots of its starting bracket; a dense unconstrained scan of the neighborhood is the arbiter of record; it relocated table entries during validation that a bracketed local refinement had placed on a neighboring zero. The in-plane double-root criterion is not sufficient: zeros exist with exact constrained/unconstrained coincidence and edge-residual ratios above 15. The EVEN spurious zero of Fig. 2 (raw $\Omega = 2.817613$) has $\log_{10}\sigma_{\min} = -4.82$, deeper than every physical flexural mode at the production basis (Table 6: -2.81 to -4.78); the residual screen rejects it at $r_M = 75.52$ (Supplementary Material, Table S.8). The residual thresholds (the per-class physical/spurious ranges quoted under instrument (i)) and the MAC forced-call threshold of §6.3 were set by inspection of the gap between physical and spurious populations in the primary tables. Their sensitivity to basis size has since been characterized directly: instrument (i) was evaluated for every validated mode and every catalogued spurious zero at eight basis sizes, $n = 16$ through 44. For the flexural problem the screen is robust: every physical mode reads 0.002–0.112 and every spurious zero 37–505 at every basis size, with no overlap, and no verdict changes when the retained branch set is independently re-derived. For the extensional problem the pointwise residual is not portable across basis sizes. Evaluated at the tabulated frequency rather than each basis's own root, physical modes can read far above their calibrated range at small n. At enlarged n the wavenumber window must be widened before the basis fills, and the search can then lock onto an adjacent branch; every such inflated residual returned to range under a denser branch derivation at the same point. The extensional screen is therefore used only at each candidate's own converged root, on the nominal window, at $n = 20/28$, with marginal cases routed to instruments (ii)–(iii).[1] For flexure,

[1] Directly confirmed at both adjudication sizes on the complete FF-P1 in-plane REAL-like/ARTIFACT-like set, evaluated at each candidate's true production-detector own root rather than a coarse-grid stand-in: $n = 20$ gives

that screen is now also unchanged at working precision dps = 25/40/60 on the primary geometry and on the two other fully-adjudicated geometries ($r_0/2b = 1.5$, $2\Theta/\pi = 1.0$ and $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$): all twelve published REAL-like/ARTIFACT-like controls keep their verdict at every dps, with r_M identical to the printed digits.[2] Figure 2 reconstructs the $\sigma_{\min}$ neighborhood of that artifact together with three physical flexural mode shapes and the EVEN spurious zero, visualizing the displacement pattern instrument (i) distinguishes.



(a) $\sigma_{\min}$ trace, raw $[2.55, 2.90]$



(b) physical, $\Omega = 0.383695$

(c) physical, $\Omega = 1.369611$

(d) physical, $\Omega = 2.602500$

(e) spurious, $\Omega = 2.817613$

Figure 2: $\sigma_{\min}$ neighborhood of the EVEN artifact (a) and reconstructed mode shapes at $n_{\text{dofs}} = 20$ (b–e). Panels (b)–(d) are physical flexural modes; (e) is the displacement-free spurious zero of §5 ($r_M = 75.52$). Unlabeled minima in (a) are out-of-table-window or rejected scan features, not additional tabulated modes.

6 Comparisons and tabulated frequencies

A fully-default detector pass is not certified complete (§4); the tables in this section use the §5 instruments. The flexural and extensional tables are not the same evidence class and are not labeled by a single word. Flexural frequencies (§6.2–§6.3) are compared to mesh-converged FE under a two-sided, portable residual screen; their only independent published check is at a different radius ratio (§6.10: 6 of 6 refined roots). Extensional frequencies (§6.3, §6.8, §6.9) are compared to FE *and* to two published semi-analytical tables, but only under a one-sided, geometry-specific adjudication: there is no portable in-plane *residual* bar. A two-sided bar on mode-shape correlation

max(REAL-like) = 0.349, min(ARTIFACT-like) = 5.199 (separation factor 14.9×); $n = 28$ gives max(REAL-like) = 0.072, min(ARTIFACT-like) = 3.769 (separation factor 52.4×), with three ARTIFACT-like seeds' own-root search correctly declining a wide, shallow lock-on and falling back to the tabulated-seed residual instead. Job-level detail: Supplementary Material, §S.4.

[2] Extensional residual ratios at the same three geometries still overlap at $r_0/2b = 2.0$, so no portable in-plane bar is claimed *on this instrument*; see §6.6, and, for a two-sided extensional bar that needs no finite-element eigenvector at all, §6.7. Job-level detail: Supplementary Material, §S.4.

is established separately in §6.6; tabulated residual ratios are unchanged. That bar is used in §6.5 to resolve eight residual-unresolved candidates at a second radius ratio, and in §6.6 on all 24 extensional keys of the geometry sweep. A second, FE-free two-sided bar (SUBDOM) is established in §6.7: it certifies a root from the boundary determinant alone. The cantilever replication (§6.1) is a third class: agreement with tables published before this implementation existed. §6.2 uses the orthotropic material of Shi, Liang, Wang and Teng [14] in an independently-run FE model, not their printed free-free row.

Table 2 maps every geometry treated below to what it was checked against and the headline result, before the subsections that follow work through each one.

Table 2: Validation roadmap: every geometry/motion pair, what it is checked against, and the headline result. FE = independently-run mesh-converged finite-element model (§6.2–§6.5); lit. = independently published table, matched directly (§6.1 and §6.8–§6.10). One-to-one = every candidate individually cross-matched to an FE mode; raw rate = nearest-neighbor FE-match fraction; MAC = the two-sided mode-shape bar of §6.6. $\nu = 0.30$ except the 24-geometry sweep and the extensional MAC sweep (both $\nu = 0.35$). *one-to-one matching does not mean every target has its own distinct root (see text).

§	R_i/R_o, $2\Theta/\pi$	Motion	Checked against	Standard	Headline result
6.1	annular cantilever (8 OOP + 3 IP geometries)	OOP+IP	[1, 2] (lit.)	one-to-one	all 29 published entries reproduced
6.2	0.5, 0.5 (orthotropic)	OOP	FE	one-to-one	8/8, max err 1.08%
6.3	0.5, 0.5 (primary)	OOP	FE	one-to-one	8/8, max err 0.64%
6.3	0.5, 0.5 (primary)	IP	FE	one-to-one*	26 roots cover 27 FE modes, worst 0.052%
6.4	0.500–0.667 × 6 angles	OOP+IP	FE	raw rate	1256/1475 (85.2%) within 3%
6.5	0.5, 1.0	OOP	FE	one-to-one	15/15 (ν=0.30) + 14/14 (ν=0.35) REAL-like
6.5	0.5, 1.0	IP	FE	raw rate	61/64 (ν=0.30), 57/61 (ν=0.35)
6.5	0.6, 1.0	OOP	FE	one-to-one	18/18 REAL-like, all FE-matched
6.5	0.6, 1.0	IP	FE	raw rate	68/75 (90.7%)
6.5	0.6, 0.5	OOP	FE	one-to-one	10/10 REAL-like, all FE-matched
6.5	0.6, 0.5	IP	FE	raw rate	43/47 (91.5%)
6.6	3 geometries + 24-key sweep	IP	FE eigenvectors	MAC	every candidate separated (0.744)
6.7	4 geometries + full blind set	IP	none (FE-free)	angle/SUBDOM	50/50 curated; 70/77 blind, 73 after 3 corrected pairings
6.8	0.5, 1.0 (orthotropic)	IP	[10] (lit.)	one-to-one	8/8, max err 0.649%
6.9	0.5, 0.5 (primary)	IP	[8] (lit.)	one-to-one*	all 6 published values accounted for
6.10	0.4, 0.5	OOP	[15] (lit.)	one-to-one	6/6 refined roots, worst 0.149%

6.1 Cantilever replication

The implementation reproduces the tabulated frequencies of [1, 2]. The solver-computed side of this comparison (Table 3, Table 4) was regenerated from scratch under the deployed `SOLVER_VERSION 2026-07-10.s10`, one geometry per task, with no value carried over from an earlier capture: every published entry matches, to 1.23% out-of-plane and 1.39% in-plane.

The largest out-of-plane residual is 1.23%, at the narrowest sector $2\Theta/\pi = 0.25$; the thin-ring row $r_0/(2b) = 5/2$, where the source papers reported slower convergence, agrees to 0.10%. The cantilever table does not contain the primary free-free geometry ($r_0/(2b) = 1.5$, $2\Theta/\pi = 0.5$); the nearest tabulated rows are $r_0/(2b) = 1.25$, $2\Theta/\pi = 0.50$ (0.28–0.90%) and $r_0/(2b) = 5/3$ (0.15%). Supplementary Table S.1 shows all eight free-free flexural labels move by less than 0.18% from $n = 20$ to $n = 36$, so the 0.64% FE residual of Table 6 is larger than the documented basis-size movement at this geometry. Several extensional labels in Table S.2 move by 0.1–0.5% with basis size, which exceeds the 0.052% FE residual of Table 7: that residual is a match to FE, not a claim

Table 3: Cantilever replication, OOP vs. [1].

$r_0/(2b)$	$2\Theta/\pi$	Mode	Computed	Paper	Err (%)
1.25	0.25	1	0.345324	0.341120	1.232
1.25	0.25	2	1.018530	1.013440	0.502
1.25	0.25	3	1.722434	1.718720	0.216
1.25	0.50	1	0.099419	0.099060	0.363
1.25	0.50	2	0.343742	0.340670	0.902
1.25	0.50	3	0.605904	0.604220	0.279
1.25	0.75	1	0.050717	0.050830	0.221
1.25	0.75	2	0.151458	0.149870	1.060
1.25	0.75	3	0.376503	0.376280	0.059
1.25	1.00	1	0.033153	0.033220	0.201
1.25	1.00	2	0.080358	0.079770	0.738
1.25	1.00	3	0.243521	0.243120	0.165
1.25	1.25	1	0.024652	0.024640	0.048
1.25	1.25	2	0.049473	0.049340	0.269
1.25	1.25	3	0.151555	0.151160	0.261
1.25	1.50	1	0.019790	0.019730	0.305
1.25	1.50	2	0.034523	0.034550	0.078
1.25	1.50	3	0.097470	0.097260	0.216
5/3	1.00	1	0.018023	0.018050	0.148
5/2	1.00	1	0.007762	0.007770	0.102

Table 4: Cantilever replication, IP vs. [2].

$r_0/(2b)$	$2\Theta/\pi$	Mode	Computed	Paper	Err (%)
1.25	0.25	1	0.341243	0.345890	1.343
1.25	0.25	2	0.808929	0.819210	1.255
1.25	0.25	3	0.940032	0.940120	0.009
1.25	0.50	1	0.120677	0.120880	0.168
1.25	0.50	2	0.347033	0.351910	1.386
1.25	0.50	3	0.529239	0.529300	0.012
1.25	1.00	1	0.039466	0.039480	0.035
1.25	1.00	2	0.106529	0.105560	0.918
1.25	1.00	3	0.263049	0.261290	0.673

that the tabulated in-plane digits have converged to 0.05%.

6.2 Orthotropic free-free (ANSYS finite-element benchmark)

The benchmark configuration is a completely free orthotropic annular sector, radius ratio $R_i/R_o = 0.5$, sector angle $2\Theta = \pi/2$, using the orthotropic material adapted from Shi, Liang, Wang and Teng [14] (§3.4). Targets are an independently-run, mesh-converged ANSYS [16] SHELL281 model of this same geometry and material ($64 \times 64/96 \times 96$ refinements, $< 0.01\%$ agreement), not values transcribed from a published table. [14] specify $E_r = 40E_\theta = 2800$ GPa; the value solved here, $E_r = 40$ GPa, is a different material, identified as such only after this benchmark was built, and no agreement with their printed free-free row is claimed. The reduction formulae of §3.4 are separately checked against a published orthotropic table in §6.8, 8 of 8 to 0.649%. All 8 targets match (Table 5).

Table 5: Orthotropic free-free flexural modes vs. ANSYS (Ω_{lit}; orthotropic material adapted from Shi, Liang, Wang and Teng [14], §3.4). Max err 1.08%; isotropic counterpart is Table 6 (0.64%). Conversion constants $T = 0.672859$, $R^* = 1.75$, $\mu_\theta = 0.525$.

FEM	Solver (Ω_{lit})	err (%)
12.40	12.5335	1.08
15.57	15.6298	0.38
36.13	36.3107	0.50
43.65	43.8826	0.53
66.76	67.2235	0.69
87.46	88.0668	0.69
89.82	90.3233	0.56
95.35	96.0312	0.71

6.3 Isotropic free-free vs. finite elements

Table 6 and Table 7 report frequency in two different conventions, not a single shared unit: the flexural table converts to the classical literature ratio Ω_{lit} (§3.1; Supplementary Material, §S.3.6) to match Shi, Liang, Wang and Teng's and the original Seok–Tiersten convention, while the extensional table reports the raw solver Ω directly against FE frequency in Hz, since Ω is already the correct in-plane ratio with no further conversion. The two columns are not directly comparable. Benchmarks are mesh-converged ANSYS models, with a half-sector symmetric/antisymmetric study certifying each benchmark mode's parity. The flexural benchmark uses SHELL281 shell elements in a two-pass deck at 64×64 and 96×96 divisions, with matched elastic modes required to agree to $< 0.3\%$ between the two meshes before use; the realized agreement is far tighter, the eight tabulated flexural modes differing by at most $1.5 \times 10^{-4}\%$ between the two passes. Because a free-free shell model interleaves the flexural, extensional, and rigid-body families in one spectrum, the numerically-zero rigid modes are discarded and extensional-family modes are identified and removed by frequency-matching against the companion plane-stress model. The extensional benchmark itself is that PLANE183 plane-stress model, built as the same two-pass $64 \to 96$ refinement and held to the same standard, with a realized worst-case difference of $5.8 \times 10^{-4}\%$ over the 27 elastic modes below 1120 Hz. Both convergence passes are contained in the released ANSYS output files, so either figure can be recomputed directly.

One systematic difference between the two sides sets a floor on the agreement reported below and is worth naming. SHELL281 is a first-order shear-deformable element; the present solution is classical Kirchhoff (§3.1). Shear flexibility and rotary inertia lower a finite-element frequency relative to a Kirchhoff one by roughly $\frac{1}{2}\left[Dk^2/(\kappa GH) + k^2H^2/12\right]$ at effective bending wavenumber k ($\kappa = 5/6$, G the shear modulus), which for this plate is 0.03% at the first tabulated flexural mode rising to 0.19% at the eighth. That is the right sign in every row — the solver sits above FE in all eight rows of Table 6 and all eight of Table 5 — and the right order of magnitude, so part of the one-sided residual reported below is a difference between two plate theories rather than error in either. The one comparison in this paper against a solution built on that same classical theory (§6.10, Shi et al.'s thin-plate Fourier-series annular sector table) agrees an order of magnitude more tightly, 0.002–0.149%, and is not one-signed.

Rigid-body modes (one transverse translation and two rotations for flexural motion; two in-plane translations and one rotation for extensional motion) occur at zero frequency for the completely free plate, lie below every scanned frequency window, and are excluded from all mode counts and matching on both the solver and FE sides: mode 1 in every table is the first elastic mode, following the literature convention. Table 6 lists the physical flexural matches, all $\leq 0.64\%$; the rejected zeros are Supplementary Material, Table S.8 (the last row is Fig. 2(e)). Extensional: 26 distinct solver roots (Table 7); 25 of the 27 extensional FE modes below 1120 Hz have a unique partner (worst 0.052%); the FE pair at 459.5/460.6 Hz is split by the solver (0.006% each) while the pair at 1114.9/1115.3 Hz is one production-detector root, so 27 FE frequencies are accounted for, not 27 independent assignments.[3] Table 6 reports raw $\bar{k}\bar{\Omega}$, not Ω; Table 7's raw column is the plain $\Omega = \omega/\bar{\omega}$ (§3.1).

The extensional count above does not include the flexural FE mode at $\Omega_{\text{lit}} = 111.46$, which lies above Table 6's window and is left as an open assignment: the solver has two ODD-parity dips, at raw 2.828013 (0.16% from the FE value) and raw 2.838515 (0.54% away). An eight-instrument

[3] An isolated refinement finds two distinct, shift-stable roots underneath this one production singularity, each within 0.015% of one FE member; MAC against both shows they are the *same* shape (one shape, two frequencies), so the single-root table entry is correct as tabulated, not a missed split. Full resolution, including the isolated frequencies and MAC values: Supplementary Material, §S.3.4.

Table 6: Free-free flexural modes vs. FEM (Ω_{lit}). Depth is $\log_{10}\sigma_{\min}$ at the n=20 screening basis of §5.

Raw $\bar{k}\bar{\Omega}$	f(Hz)	Solver (Ω_{lit})	FEM	err (%)	depth
0.383695	4.732	15.148	15.128	0.13	−3.42
0.616645	7.605	24.344	24.190	0.64	−3.91
0.961312	11.855	37.951	37.818	0.35	−3.69
1.369611	16.890	54.070	53.828	0.45	−4.78
1.752188	21.608	69.174	68.847	0.47	−3.91
2.320741	28.620	91.619	91.452	0.18	−2.81
2.379886	29.349	93.954	93.537	0.45	−4.21
2.602500	32.095	102.743	102.136	0.59	−3.70

Table 7: Free-free extensional modes: raw Ω vs. FEM. Twenty-six distinct solver roots; 25 FE modes have a unique partner. Entry 26 also matches the 1115.3 Hz partner, so 27 FE frequencies are accounted for, not 27 independent assignments (§6.3). Raw column is not comparable to Table 6 (§3.1).

#	Ω	FEM f(Hz)	err(%)	#	Ω	FEM f(Hz)	err(%)
1	0.401154	161.4	0.025	14	1.832694	736.8	0.052
2	0.687446	276.5	0.007	15	1.972966	793.6	0.001
3	0.764697	307.5	0.030	16	1.987180	799.3	0.003
4	1.142287	459.5	0.006	17	1.996235	803.0	0.004
5	1.145158	460.6	0.006	18	2.166815	871.6	0.002
6	1.197600	481.7	0.005	19	2.293214	922.3	0.013
7	1.416689	569.7	0.026	20	2.301512	925.7	0.007
8	1.440568	579.3	0.027	21	2.389113	960.7	0.031
9	1.522330	612.3	0.007	22	2.450795	985.7	0.011
10	1.598985	643.1	0.012	23	2.521489	1014.1	0.014
11	1.613123	648.8	0.010	24	2.566362	1032.0	0.029
12	1.690020	679.8	0.001	25	2.676026	1075.9	0.047
13	1.779792	715.9	0.001	26	2.771955	1114.9	0.008

Raw column is $\Omega = \omega/\bar{\omega}$, not $\bar{k}\bar{\Omega}$; not comparable to Table 6 (§3.1).

suite fails to force an assignment between them; the individual instruments and their outcomes are in Supplementary Material, §S.3.2. The decisive instrument is MAC: alone among the eight, it judges mode *shape* instead of frequency or matrix structure. Both candidates reconstruct as genuinely ODD-parity and correlate almost perfectly with the FE eigenvector (cand. 2.828013: MAC = 0.9913; cand. 2.838515: MAC = 0.9999), a gap of only 0.0086 against the 0.2 forced-call threshold used here, more conservative than the conventional experimental-modal bands of Ref. [17] (0.9/0.05 good/uncorrelated). Two further, independent checks (basis-size stability and an independent pure-slope-residual comparison; Supplementary Material, §S.3.2) confirm both are radially distinct, genuine modes, not one a numerical echo of the other. Both are therefore confirmed genuine, high-fidelity ODD-parity modes; the FE mode is not missing from the solver spectrum, and the pair is best interpreted as a real, basis-independent near-degenerate pair ($\Delta\Omega \approx 0.0105$, $\sim$0.4% apart) whose mode shapes are too similar for any instrument to separate; the assignment is left unforced, not resolved. Candidate 2.828013's self-parity is basis-sensitive beyond the calibrated range (Supplementary Material, §S.3.2).

6.4 Geometry generalization: radius ratio and sector angle sweep

Tables 6 and 7 validate the free-free detector at a single geometry. To check how broadly that agreement generalizes, a second, independent sweep was run over 4 radius ratios ($R_i/R_o = 0.500$–0.667) × 6 sector angles ($\Theta = 22.5°$–$135°$) — 24 geometries, each run for both motion types, so 48 geometry/motion combinations — each run through the full solver pipeline against an independent FE benchmark (a matching pair of ANSYS SHELL281/ PLANE183 models per geometry, same two-mesh convergence standard as §6.2) at a second isotropic material ($\nu = 0.35$, so this is a generalization check against its own FE ground truth, not a re-validation of the $\nu = 0.30$ geometry above).

Matching is nearest-neighbor, a coarser adjudication than Tables 6–7's one-to-one, parity-checked assignment: no per-mode FE partner is claimed. The §5 residual screen was applied to every accepted flexural candidate (695 targets across the 24 OOP geometry keys)[4]: 654 reconstructed at the production basis, 329 REAL-like / 321 ARTIFACT-like / 4 AMBIGUOUS. The REAL-like fraction is 50.3% over all 654 — 47.9% over the 119 cut-off-adjacent candidates among them against 50.8% over the other 535 — so the calibrated flexural bar travels and does not degrade near the cut-off. The 41 reconstruction failures are the known even-valued basis-fill shortfall, 33 of them at $r_0/2b = 2.5$, $2\Theta/\pi \leq 0.5$. Extensional candidates have no portable verdict bar on the residual (§5, §6.5); the mode-shape bar of §6.6 was applied to all 24 extensional keys of this sweep at the sweep material $\nu = 0.35$. Setting aside a known cut-off-adjacent subset with elevated error (consistent with the near-cutoff conditioning behavior already noted in §5), agreement is close and geometry-independent: 1256 of 1475 candidates (85.2%) matched an independent FE mode within 3%, mean of per-geometry mean errors 0.574%, with no degradation trend as R_i/R_o increases (Supplementary Material, Table S.3). The 3% window measures coverage, not precision, and on a spectrum this dense it is loose: a frequency drawn uniformly across a geometry/motion pair's own FE span lands within 3% of *some* FE mode about 60% of the time, so the hit rate alone sits only modestly above chance.[5] What is not near chance is how close the matches are: matched candidates' errors have median 0.315%, with 63.5% inside 0.5% and 77.7% inside 1%, against a median near 1.5% for a uniform draw in the same window. The precision of the agreement, not its coverage, is what this sweep shows. The per-candidate set churned substantially under the completeness work between checkpoints; 85.2% is a re-derivation of the aggregate, not an unchanged candidate list.[6] This does not replace Tables 6–7 as the primary tabulated comparison. Coarse nearest-neighbor agreement is geometry-independent; full residual-screened adjudication has been performed at four geometries.

6.5 Mode-by-mode adjudication at three further geometries

The sweep above is nearest-neighbor against FE and uses $\nu = 0.35$. The mode-by-mode standard of Tables 6 and 7 is applied here at three further geometries that close the $\{R_i/R_o = 0.5, 0.6\} \times \{2\Theta = \pi/2, \pi\}$ rectangle at $\nu = 0.30$ (the first of them also at $\nu = 0.35$), each through the same default production pipeline, one-to-one FE cross-match, and §5 residual/parity screen. Per-mode outcomes are Supplementary Material, Tables S.4–S.6.

R_i/R_o, $2\Theta/\pi$	OOP REAL-like	IP (raw)	distinctive finding
0.5, 1.0	15/15 (ν=0.30) + 14/14 (ν=0.35)	61/64, 57/61	unmatched flexural candidates ARTIFACT-like only
0.6, 1.0	18/18, all $\leq 0.62\%$	68/75	residual populations overlap; FE#23 unmatched
0.6, 0.5	10/10, all $\leq 0.58\%$	43/47	those 10 cover every in-range elastic FE mode

At every geometry, every REAL-like flexural candidate matched an independent FE mode: no catalogued spurious zero passed the screen at any of the three. The converse does not hold everywhere — one in-range FE mode at (0.6, 1.0) has no REAL-like partner, below — so what these

[4]Job-level detail: Supplementary Material, §S.3.3.

[5]Monte-Carlo estimate over each pair's own FE span, using the FE modes that appear as a match in the shipped consolidated manifest — 889 of the 1517 modes actually extracted, so the true chance rate is higher still.

[6]Superseded intermediate-recapture figures and full provenance: Supplementary Material, §S.3.3, §S.4.

geometries establish is that the screen's REAL-like calls are reliable, not that its coverage of the FE spectrum is complete. Errors again fall on one side only, the same pattern as Table 6. Unmatched flexural candidates are ARTIFACT-like, concentrated in near-degenerate neighborhoods of the kind documented for the ODD pair (§6.3), and are not force-assigned.

The first geometry isolates sector angle at the primary radius ratio ($R_i/R_o = 0.5$, $2\Theta/\pi = 1.0$, a half-annulus against the primary quarter-annulus). Raising the FE extract from 35 to 80 modes recovered every extensional candidate that had sat above the old ∼798 Hz ceiling; the few remaining non-matches all lie below that ceiling and are residual-classified ARTIFACT-like or AMBIGUOUS.

The second isolates radius ratio at that new angle ($R_i/R_o = 0.6$, $2\Theta/\pi = 1.0$). One in-range FE mode has no production-capture partner (FE mode 23 of 34, $\Omega_{\text{lit}} = 129.650$); FE mode 9 ($\Omega_{\text{lit}} = 16.100$) is a second root next to the FE#10 partner, missed at n=20 because the dip sat just short of the -3.5 accept floor.[7] Neither disturbs the property above: every REAL-like call at this geometry still has its own FE partner. This geometry also shows why no extensional *residual* bar is claimed. Restricting to the 74 candidates below the FE extraction ceiling, and taking a match within 0.5% as presumptive-physical and the absence of any match within 3% as presumptive-spurious, the two populations do not separate: $r_{\theta\theta} = 0.009$–7.6 against 2.8–9.6. None of the presumptive-spurious candidates falls below 0.349, the largest REAL-like reading on the primary geometry's complete set when every candidate is evaluated at its own converged root (§5; that same set read at its catalogued seeds on the owning block reaches 0.6858, Supplementary Material, Table S.7), while 34 of 49 presumptive-physical ones do: a low residual is evidence of a physical mode here, a high one is not evidence against. Eight candidates match an FE mode tightly while reading above the spurious group's lower edge; scored on the mode-shape bar of §6.6, six are ARTIFACT-like (same-class MAC 0.18–0.58) and two are REAL-like ($\Omega = 1.667187$ against FE#33, MAC 0.9981; $\Omega = 1.894176$ against FE#42, MAC 0.9953). For four of the six artifacts the nearest-frequency FE target belongs to the opposite symmetry class, so MAC against that target is identically zero. A tight frequency match at this geometry is not by itself evidence of a physical mode.

The third geometry crosses the two axes ($R_i/R_o = 0.6$, $2\Theta/\pi = 0.5$). The ten REAL-like roots cover every in-range elastic FE mode (FE modes 7–16 of 34). FE mode 16 is also claimed by an ARTIFACT-like companion ($\Omega = 2.412108$, 1.86%), the same near-degenerate pairing as §6.3, not force-assigned. Unlike the second geometry, no in-range flexural FE mode lacks a REAL-like partner. The four extensional misses are residual-classified ARTIFACT-like.

That closes the $\{R_i/R_o = 0.5,\, 0.6\} \times \{2\Theta = \pi/2,\, \pi\}$ rectangle at $\nu = 0.30$. It does not make the 24-geometry sweep one-to-one, and it does not supply a portable extensional bar on the residual; for a two-sided bar on mode-shape correlation see §6.6.

6.6 A portable two-sided extensional bar on mode-shape correlation

Every tabulated extensional frequency above is still residual-adjudicated: no portable two-sided bar exists on the pointwise edge residual, and none is claimed. A different instrument does supply one. The eight candidates of §6.5 are scored on it. The modal assurance criterion (MAC) already used in §6.3 to resolve the flexural ceiling ambiguity draws on mode *shapes* rather than boundary residuals, so it is structurally independent of the quantity that fails to port. Throughout this subsection MAC is the identity-weighted correlation of the stacked nodal in-plane displacement (U_X, U_Y), not a mass-weighted form: if $\mathbf{a}$ and $\mathbf{b}$ are the two stacked vectors, $\text{MAC} = |\mathbf{a}\cdot\mathbf{b}|^2/(\|\mathbf{a}\|^2\|\mathbf{b}\|^2)$. Production

[7] At n=28 and n=36, $\Omega^* = 0.261289$ ($\Omega_{\text{lit}} = 16.1176$, err 0.109%, depths -6.74 and -6.43, shift 0.0002%, 2.17% from the claimed neighbor 0.267076). FE#23 had no second minimum in the unconstrained scan. Job-level detail: Supplementary Material, §S.4.

scoring takes the maximum of that quantity over FE modes of the same symmetry class as the root-owning block. The production call is the midpoint 0.744 of the $\nu = 0.30$ calibration window below. Building **a** requires evaluating the exact reconstruction at each FE node's coordinates and resolving the two models' angular origins; Supplementary Material, §S.3.2 gives that procedure, including the mesh-pairing step it depends on for classifying FE modes by symmetry (exact at $2\Theta = \pi$; a Cartesian-midside fallback at every other sector angle, including the primary $2\Theta = \pi/2$). A MAC taken across parity blocks is analytically zero, so each candidate is scored on the block that owns a root at its own Ω (Supplementary Material, §S.3.2).

Scored that way, MAC separates the two populations at all three fully-adjudicated geometries, on the same already-published REAL-like / ARTIFACT-like calibration sets used elsewhere in this paper: eight candidates per geometry, four already labeled REAL-like and four ARTIFACT-like by the residual screen or (for $r_0/2b = 2.0$) by prior one-to-one FE adjudication. At the primary geometry the REAL-like half is drawn from Table 7's own published roots; the ARTIFACT-like half are catalogued spurious zeros of the same determinant, now listed with their owning-block residuals in Supplementary Material, Table S.7 (the in-plane analogue of Table S.8). The two non-primary geometries' calibration sets are their own independently adjudicated candidates, not rows of Table 7, which covers only $r_0/2b = 1.5$, $2\Theta/\pi = 0.5$.

geometry ($r_0/2b$, $2\Theta/\pi$)	min($\mathrm{MAC}_{\mathrm{REAL}}$)	max($\mathrm{MAC}_{\mathrm{ART}}$)	gap
(1.5, 0.5)	0.9079	0.0067	+0.9012
(2.0, 1.0)	0.9869	0.5801	+0.4068
(2.0, 0.5)	0.9994	0.3480	+0.6514

Combined, with every candidate evaluated at its catalogued frequency, the REAL floor is 0.9079 and the ARTIFACT ceiling 0.5801: any threshold between them classifies all 24 candidates correctly.[8] Local-minimum block selection, with residual tie-break, classifies 24 of 24; either simpler criterion alone is 23 of 24 (Supplementary Material, §S.3.2). Two candidates held out of the calibration, where residual and FE disagree, both read MAC 0.998 and agree with the FE evidence.

Three limits bound the $\nu = 0.30$ calibration. The bar is calibrated on FE-matched Table 7 roots and independently adjudicated ARTIFACT-like zeros (one of the four primary REAL seeds, row 23, sits in the residual middle band; Supplementary Material, Table S.7), so its demonstrated added value is the tie-break above rather than a blind re-adjudication. At (2.0, 0.5) all three selection criteria score 8 of 8, so that geometry confirms the bar transfers but supplies no evidence that the rule is better than the simpler alternatives; the rule's margin rests on two candidates. The ARTIFACT ceiling on that calibration set is strongly geometry-dependent (0.0067, 0.3480, 0.5801). It is now the production two-sided extensional classification instrument when FE eigenvectors are available: local-minimum block selection, MAC scored as the maximum over FE modes of the same symmetry class as the root-owning block, threshold 0.744. It is not inside the frequency detector and does not change a computed Ω. Tabulated residual ratios are unchanged. §6.5 reports the bar's call on the eight residual-unresolved candidates at that geometry: six ARTIFACT-like, two REAL-like.

The $\nu = 0.30$ window $(0.5801, 0.9079)$ is a calibration result; nothing here claims it as a universal floor. The same production call was then applied to every accepted extensional candidate at all 24 keys of the §6.4 sweep, at that sweep's own material $\nu = 0.35$, against an independent PLANE183 extract of the same 57 elastic modes (FE modes 4–60) used at the primary calibration geometry.

[8] The three-geometry calibration table was independently reproduced bit-for-bit against a transcribed gate table, and each geometry reproduces its own previously-published per-block residuals to 10^{-4} relative (job-level detail: Supplementary Material, §S.4). Robustness of the reported window to the choice of seed-versus-own-root evaluation and to the FE-target pairing rule is established in Supplementary Material, §S.3.2.

The C-class rows at $2\Theta/\pi = 1.25$ and 1.50 have no tight MAC-REAL in that extract; repeating those eight keys against an 80-elastic-mode extract still produced none (Supplementary Material, §S.4).[9] Combined, all 1596 extensional candidates the sweep produces reconstructed (§6.4's 1475 is the smaller subset left after removing cut-off-adjacent and beyond-FE-range candidates from both motion types; Supplementary Material, Table S.3); 912 match an FE mode within 0.5%, of which 410 are MAC-REAL and 502 are MAC-ART at threshold 0.744, with no tight misclassification on either side of that threshold. The call splits by sector angle, not by radius ratio. Each row below pools that angle's four radius ratios, and its outcome column is the declared-before-pooling verdict class for the angle: A where both populations are present and the threshold separates them, C where the sweep produced no tight MAC-REAL candidate at any radius ratio.

$2\Theta/\pi$	recon.	tight	REAL/ART	$\min(\mathrm{MAC}_{\mathrm{REAL}})$	$\max(\mathrm{MAC}_{\mathrm{ART}})$	outcome
0.25	97/97	62	51/11	0.9547	0.6551	A
0.50	163/163	101	78/23	0.8218	0.6650	A
0.75	256/256	160	131/29	0.9260	0.5130	A
1.00	303/303	187	150/37	0.8495	0.6803	A
1.25	376/376	214	0/214	—	0.5993	C
1.50	401/401	188	0/188	—	0.4295	C

The sixteen keys with $2\Theta/\pi \leq 1.0$ each have at least one tight MAC-REAL and one tight MAC-ART. The eight keys at $2\Theta/\pi = 1.25$ and 1.50 (every radius ratio) have none: every tight FE-frequency match is MAC-ART, the same pattern seen among the eight residual-unresolved candidates of §6.5. Because the two C rows carry no REAL-side evidence, 0.744 was left where the $\nu = 0.30$ calibration had put it. Combined REAL floor on the sixteen keys that have one is 0.8218 ($r_0/2b = 2.0$, $2\Theta/\pi = 0.5$); combined ARTIFACT ceiling is 0.6803 ($r_0/2b = 2.5$, $2\Theta/\pi = 1.0$). Both sit inside the $\nu = 0.30$ calibration window: that window does not survive a change of Poisson ratio, radius ratio, or sector angle, and the production midpoint was not retuned. What travels is the call at 0.744, which still splits every tight candidate on all 24 keys. The rectangular companion [18] reuses that same unre-tuned call as a conservative shape-confirmation threshold on flexural half-model eigenvectors; it is not a recalibration. So a two-sided extensional bar does exist, with a measured reach; it is simply not carried by the pointwise residual.

Figure 3 makes both tables above visible at a glance: every calibration and sweep candidate's own MAC value against the production threshold, rather than only the extrema. The two populations occupy disjoint bands at every geometry; the closest approach from either side is the $2\Theta/\pi = 1.0$ sweep row's ARTIFACT ceiling of 0.6803, still 0.064 clear of the threshold.

6.7 A finite-element-free two-sided extensional certification criterion

The mode-shape bar of §6.6 closes the extensional two-sidedness gap only where an FE eigenvector is available. A second, independent instrument closes the same gap without one, certifying an extensional root as real or spurious from the boundary determinant alone, the way the flexural screen already does.

Construction. This subsection introduces a small linear-algebra notation; to keep it clear of the physical symbols of §3, its matrices are set in bold ($\mathbf{D}, \mathbf{T}, \mathbf{M}, \mathbf{R}$, distinct from the scalars D, T, $M_{r\theta}$, R^*), its traction-to-displacement ratio is η (never ρ, which is mass density throughout), and its pooling factor is γ (never G, which denotes a shear modulus elsewhere in this paper). For

[9] Tight means an FE match within 0.5%. Flexure on this sweep remains residual-screened; this is the extensional MAC application, not a one-to-one residual adjudication of the 24-geometry table.

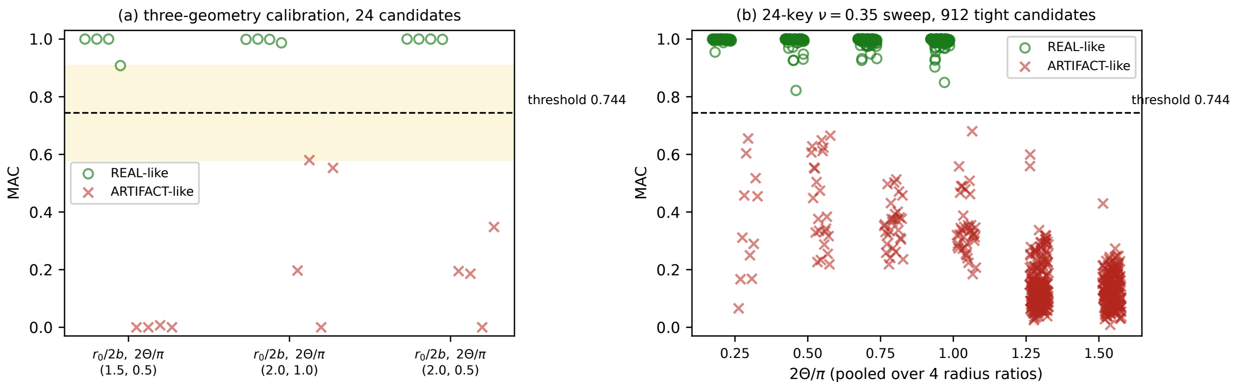


Figure 3: MAC separation of the two-sided extensional bar (§6.6), REAL-like (green circles) vs. ARTIFACT-like (red crosses) against the production threshold 0.744 (dashed). (a) The three fully-adjudicated calibration geometries of the (min REAL, max ARTIFACT, gap) table above, every one of the 24 candidates individually (not only the tabulated floor/ceiling); the shaded band is the combined ν=0.30 window $(0.5801, 0.9079)$. (b) The 24-key ν=0.35 geometry sweep of §6.4, pooled by sector angle as in the angle-pooled table above, every tight (FE match $< 0.5\%$) candidate individually; the $2\Theta/\pi = 1.25, 1.50$ keys have no REAL-like tight candidate at all (§6.6), hence no green points in those two columns.

extensional (in-plane) motion the same boundary determinant that supplies $\mathbf{K}(\Omega)$ admits an exact factorization $\mathbf{K} = \mathbf{D}^\mathsf{T}\mathbf{T}$, where $\mathbf{D}$ collects the displacement weights and $\mathbf{T}$ the traction weights of the underlying two-edge functional; the factorization was verified entrywise against the deployed assembly (gate G1, bar 10^{-30}, measured 1.21×10^{-41} to 8.80×10^{-41}). Define $\eta(c) = \|\mathbf{T}c\|/\|\mathbf{D}c\|$ for a coefficient vector c, and $\eta(\Omega) = \min_c \eta(c)$, the smallest traction-to-displacement ratio attainable at that frequency. The deployed pointwise residual screen of §5 evaluates this ratio at one particular point, the null vector of the accepted zero, $\eta(c_\text{null})$; since c_null is one feasible choice among all c, $\eta(c_\text{null}) \geq \eta(\Omega)$ identically (gate G3), which is the algebraic source of that screen's already-reported one-sidedness for extensional motion (§5): a small $\eta(c_\text{null})$ certifies a physical root, but a large one does not certify an artifact, because $\eta(\Omega)$ itself may still be small.

$\eta(\Omega)$ as a magnitude does not repair this: at the primary geometry the REAL and ARTIFACT populations of $\eta(\Omega)$ overlap, and at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$ the overlap is total. What separates the two populations is not the minimizing ratio's *size* but an *angle*: at a genuine root, $\mathbf{K}$'s null direction and the traction-minimizing direction are the same physical displacement field; at a projection artifact they are unrelated vectors that happen to share a small determinant. Two refinements make that angle a working instrument. First, near-degenerate directions must be pooled rather than compared one at a time: with $\sigma_1 \leq \sigma_2 \leq \cdots$ the generalized singular values of the whitened pencil ($\mathbf{D} = \mathbf{QR}$, $\mathbf{M} = \mathbf{TR}^{-1}$, SVD of $\mathbf{M}$), the criterion projects onto the cluster of the $p = \#\{i : \sigma_i \leq \gamma\sigma_1\}$ lowest-traction directions, with $\gamma = 4$ the pooling value used in every result below. $\gamma = 3.0$ was the original pre-registered choice; the labelled sets already showed the REAL/ARTIFACT separation stable across $\gamma = 1.5$–8 before any blind run, and a second-geometry blind set is inert for $\gamma = 2$–6. The one $\gamma = 3.0$ miss in the first blind geometry (FE mode 23 below) sits at a knife-edge $\gamma = 3.01$; $\gamma = 4$ is a round value inside that pre-established window, not a fit to 3.02. Second, the projection must be taken on the *interior* displacement field, not the edge trace: an edge-trace version of the same angle is defeated by a specific artifact (a spurious mode-2 component that "bleeds through" the edge trace but not the interior field, reading 0.8692 on the edge against 0.1426 in the interior for the same candidate).

SUBDOM is the fraction of the null vector's own interior displacement-field energy that lies in the span of the p lowest-traction directions' own interior fields: the p candidate fields are computed on the interior $r \times \theta$ grid, Gram–Schmidt orthonormalized under the same identity-weighted inner product used for the §6.6 MAC, and SUBDOM is the cumulative squared projection of the (normalized) null field onto that orthonormal span. **The call is SUBDOM $> 0.70 \Rightarrow$ REAL**, set as the midpoint of the labelled-set window below and then frozen: it was fixed before any cluster run — the four-geometry curated confirmation and the 122-candidate blind set alike — and never retuned afterward. "Pre-registered" below refers to that freeze. Six gates guard the computation: G1 verifies the $\mathbf{K} = \mathbf{D}^\mathsf{T}\mathbf{T}$ factorization entrywise; G2 reproduces the published FF-P1 extensional residual ratios (10^{-4} bar, fires only at the primary geometry/basis); G3 is the algebraic identity $\eta(\Omega) \leq \eta(c_{\text{null}})$; G4 cross-checks the edge-trace decomposition (SUB1) against an explicit MAC computation (10^{-9} bar; SUB1 is not SUBDOM, and the two are supposed to differ, since one is edge and the other interior); G5 confirms the candidate is evaluated at a genuine local minimum of σ_{min}; G6 is a negative-control siting check, not applicable in blind mode. Every applicable gate passed at every geometry reported below (G2 fires only at the primary geometry and production basis; G6 is not applicable in blind mode). Supplementary Material, §S.3.7 gives the numerical procedure (whitening, gate bars, calibration set, and the inherited block-selection rule) at reimplementation detail; job numbers for the runs below are §S.4.

Calibration. At $n_{\text{dofs}} = 20$, across four (geometry, ν) combinations, 52 labelled points (two historical "killer" near-degenerate modes and 13 non-root negative controls included) place the REAL floor and ARTIFACT ceiling in a combined window of $(0.6091, 0.7904)$, all 52 correctly classified. The threshold is that window's midpoint, $(0.6091+0.7904)/2 = 0.6998$, rounded to 0.70, rather than either endpoint; everything below is scored at that value, unchanged.

Cluster confirmation. The criterion was then run, unmodified and gate-verified, on the SLURM cluster rather than only in the diagnostic container. On curated candidate lists (13–15 hand-labelled REAL, ARTIFACT, held-out and negative-control points per geometry) at all four of this paper's standard geometries, every geometry scores every candidate correctly at the pre-registered threshold, with zero cross-population overlap:

geometry ($r_0/2b$, $2\Theta/\pi$; ν)	scored	REAL floor	ARTIFACT ceiling	window vs. §6.6's 0.407
(1.5, 0.5; 0.30)	11/11	0.9901	0.4668	+0.5233, wider
(2.0, 1.0; 0.30)	15/15	0.9899	0.5665	+0.4234, wider
(2.0, 0.5; 0.30)	12/12	0.9990	0.5587	+0.4403, wider
(1.5, 1.0; 0.35)	12/12	0.9998	0.6091	+0.3907, narrower

Three of the four windows are wider than §6.6's own 0.407[10] FE-dependent gap; the $\nu = 0.35$ geometry's window is narrower: the margin is wider at three geometries and narrower at one, not uniformly larger everywhere.[11]

A harder test scores every candidate in a geometry's complete adjudication set, with no ground-truth label of any kind entering the SUBDOM computation; truth is assigned afterward, from each

[10]The narrowest of §6.6's three per-geometry gaps, at (2.0, 1.0) ($0.9869 - 0.5801 = 0.4068$, using each candidate's own converged root rather than its catalogued seed), not §6.6's combined three-geometry window (REAL floor 0.9079, ARTIFACT ceiling 0.5801, gap 0.3278). Full derivation: Supplementary Material, §S.3.2.

[11]An independent mode-shape (MAC) cross-check against ANSYS eigenvectors, unavailable for this fourth geometry when this calibration was first run, has since been completed at all 12 of its labelled points (Supplementary Material, §S.3.7 and §S.4): every REAL-labelled point reads MAC > 0.999 on the SUBDOM-selected block and every ARTIFACT-labelled point rejects on that same block, confirming the labels used above by a second, structurally independent instrument. This is a diagnostic cross-check, not a recalibration; the numbers in the table are unchanged.

candidate's own independent FE nearest-frequency match. Run as an array of independent SLURM tasks and merged by a script that recomputes the verdict from each row's own stored gate record (so the merged answer does not depend on task order), this blind test at $\gamma = 4$ resolves 45 of 49 FE modes at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$ and 25 of 28 at $r_0/2b = 2.0$, $2\Theta/\pi = 0.5$, with **zero duplicate zeros ever accepted** at either geometry: whenever a tighter and a looser frequency claimant both exist for one FE mode, SUBDOM never scores the looser one above threshold. (At the original $\gamma = 3.0$ the same stored rows score 44 of 49 and 25 of 28; the extra resolution is FE mode 23 at $\Omega = 1.411930$, a knife-edge $\gamma = 3.01$ where the pointwise residual is REAL-like and SUBDOM at $\gamma = 3.0$ is not.) Of the seven first-pass misses at $\gamma = 4$ against nearest-frequency matching, the failures are not one kind. Three (FE modes 35, 50 and 53 at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$) are pairing defects: the frequency-nearest solver candidate is not the FE mode shape, but a nearby SUBDOM-real structure is, confirmed independently below; those three raise the $\gamma = 4$ score from 45 to 48 of 49. Two more (FE modes 16 and 30 at $r_0/2b = 2.0$, $2\Theta/\pi = 0.5$) are pairing artifacts of the same kind with no replacement root found. Two are block-selection misses of the inherited local-minimum-residual rule: FE mode 32 at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$ (auto-picked block MAC = 0.0000; the other block matches at 0.8528 and SUBDOM = 0.7136) and FE mode 26 at $r_0/2b = 2.0$, $2\Theta/\pi = 0.5$ (MAC = 0.9302 and SUBDOM = 0.8728 on the non-picked block). FE mode 26 recovers at an enlarged basis, treated below; FE mode 32 does not.

Three FE nearest-frequency pairing defects, confirmed independently. At $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$, three FE modes (35, 50 and 53) each sit within $\lesssim 0.01$ in Ω of two distinct solver structures: the frequency-nearest candidate from the blind catalogue, and a second, previously uncatalogued SUBDOM-real structure nearby. Ordinary nearest-frequency FE matching cannot separate two roots this close, since proximity in frequency does not imply similarity in mode shape. In each case an ANSYS extraction of the FE mode's own eigenvector, compared by identity-weighted MAC against both candidate mode shapes (each reconstructed from its own explicitly forced parity block), confirms the SUBDOM-selected structure and rejects the frequency-nearest one.

FE mode 35 (682.741 Hz) is the worked example: the criterion's own prediction — that the FE mode belongs to the SUBDOM-real structure at $\Omega = 1.697604$ (SUBDOM = 0.988) rather than the sharp true zero at $\Omega = 1.698604$ (SUBDOM = 0.097) — was logged before any FE cross-check and then confirmed: MAC = 0.9944 against the SUBDOM-picked structure and MAC = 0.0000 against the frequency-nearest one (ANSYS mesh64/mesh96 cross-converged at MAC 0.9991), agreeing with the independent traction residual (0.186, REAL-range, against 5.378, ARTIFACT-range). Table 8 gives the same resolution for FE modes 50 and 53. FE mode 50's cross-check used an independent frequency-domain route as well: converting each candidate Ω to Hz via a tightly FE-matched neighbouring mode (FE mode 49, 0.0058% error) puts the SUBDOM-real structure at 862.44 Hz, 0.0045% from FE mode 50's own ANSYS-extracted value (862.40369 Hz), against 0.108% off for the catalogued point. FE mode 53's replacement root needed a different polishing method: the two parity blocks' $\sigma_{\min}$ cross inside the candidate's own ± 0.001 local-minimum window, so the standard unconstrained golden-section primitive locks onto a deeper, non-matching-block feature instead of the true root; a block-constrained variant, minimising only the matching block's own $\sigma_{\min}$ in isolation, converged cleanly to $\Omega^* = 2.265896$, 2.8 decades deeper on the matching block than the non-matching one there.

Correcting all three pairings raises $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$'s $\gamma = 4$ blind-set resolution from 45 to 48 of 49 FE modes against FE nearest-frequency matching; combined over both geometries, the blind test at $\gamma = 4$, scored against FE nearest-frequency matching and the inherited block rule and counting these three pairings, resolves 73 of 77 FE modes across 122 candidates[12] with zero duplicates ever accepted.

Table 8: Three FE nearest-frequency pairing defects at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$: the catalogued (frequency-nearest) candidate is rejected by MAC and SUBDOM; the SUBDOM-real replacement is confirmed by MAC against the FE eigenvector. Job-level detail: Supplementary Material, §S.4.

FE mode (f, Hz)	catalogued Ω	$\mathrm{SUBDOM}_{\mathrm{cat}}$	$\mathrm{MAC}_{\mathrm{cat}}$	replacement Ω	$\mathrm{SUBDOM}_{\mathrm{new}}$ / $\mathrm{MAC}_{\mathrm{new}}$
35 (682.741)	1.698604	0.097	0.0000	1.697604	0.988 / 0.9944
50 (862.404)	2.146515	0.0894/0.4813	0.0000	2.144291	0.9974 / 0.9993
53 (911.050)	2.259637	—*	0.2310*	2.265896	0.9965† / 0.9978

*best MAC on the matching parity block, below the 0.744 threshold; SUBDOM was not separately gated at the catalogued point. †SUBDOM at the pre-polish scan point $\Omega = 2.265637$; the block-constrained golden-section polish (text) converged to $\Omega^* = 2.265896$.

FE mode 26: a block-selection miss recovered at an enlarged basis. FE mode 26 (catalogue $\Omega = 2.068435$ at $r_0/2b = 2.0$, $2\Theta/\pi = 0.5$; MAC = 0.9302/SUBDOM = 0.8728 on the non-picked block at the production basis) is not a same-basis pairing defect like FE modes 35, 50 and 53: at $n_{\mathrm{dofs}} = 20$, $x_{\max} = 20$ the FE-matching parity block is never a local minimum at all, so no rule gated on local-minimum qualification can reach it (a stored-data-only proof, no new solves, that also rules out max-over-blocks, which accepts duplicate zeros on the blind set and is not adopted). A cluster scan across two enlarged bases ($n_{\mathrm{dofs}} = 24$, $x_{\max} = 40$ and $n_{\mathrm{dofs}} = 28$, $x_{\max} = 60$) instead relocates the root: at the larger basis, a single continuous $\sigma_{\min}$ valley — confirmed by a finer connecting scan that rules out a coincidentally adjacent second well — has its true minimum at $\Omega = 2.110435$ (scan grid), polished by the same golden-section refinement every other catalogued Ω receives to $\Omega = 2.110861$ (fidelity gate against an already-validated same-basis catalogue root: drift 1.1×10^{-3}, inside the 5×10^{-3} bar). There the inherited block rule cleanly picks the FE-matching block via a single qualifying local minimum (SUBDOM = 0.9991), and an ANSYS mesh96 eigenvector extraction for FE mode 26 confirms it independently: MAC = 0.8906, comfortably above the 0.744 bar. This is a basis-enlargement recovery, not a same-basis correction: it holds at $n_{\mathrm{dofs}} = 28$, $x_{\max} = 60$ and is not counted in the $n_{\mathrm{dofs}} = 20$ production-basis blind-set score above, which is basis-specific by construction. FE mode 32, the other block-rule miss, does not recover the same way: at both enlarged bases tested its relocated well is an unrelated mode (MAC ≈ 0.06 against FE mode 32's own eigenvector), so it stays an open miss.

What this does not yet establish. Basis portability at an enlarged basis is now cluster-tested. At $n_{\mathrm{dofs}} = 24$, $x_{\max} = 40$ the labelled local-minimum points still separate at all four geometries (the in-container r200/a100 window is reproduced: REAL 0.9418–1.0000 versus ARTIFACT 0.3779–0.5687). At $n_{\mathrm{dofs}} = 28$, $x_{\max} = 60$ many catalogue Ω are no longer roots (`C_MOVED`, excluded); among those that remain, the historical near-degenerate killer at $\Omega = 1.972966$ is still a local minimum with SUBDOM = 0.1857 (a characterised miss, not a moved root), while a certifiable well 0.009 above it, at $\Omega = 1.981966$, is the neighbouring genuine root 1.987180 relocated at this basis. A full blind re-score at that basis resolves 28/31 and 14/15 FE modes that still have a local minimum, with zero duplicates accepted. Block selection at a near-degenerate pair still uses the same local-minimum-residual rule inherited from §6.6's own production pipeline, itself a

[12] 75 candidates at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$ (job 2445957) and 47 at $r_0/2b = 2.0$, $2\Theta/\pi = 0.5$ (job 2445958); Supplementary Material, §S.4.

form of automatic singular-value-based selection — the same style of mechanism whose blind trust produced the pairing-defect probe's own first-attempt bug (an exact 0.0000/0.0000 result later traced to both candidates silently reconstructing the same block), caught before being reported and fixed by forcing the block explicitly, with an in-run fidelity gate proving the fix reproduces the original unforced behavior bit-for-bit. Forcing both blocks at the remaining disputed modes shows that this inherited rule, not the interior projection, is what misses FE mode 32 (FE mode 26 recovers at an enlarged basis, above). The rule is unmodified. Scoring max-over-blocks instead accepts duplicate zeros on the $n = 20$ blind set, so that alternative is not adopted. FE mode 30's non-picked block is a real field: it matches FE mode 28 at MAC = 0.7709, not FE mode 30. The $\nu = 0.35$ geometry's own ground truth remains this project's construction rather than a published set. Working precision was checked on the full labelled set at all four geometries (dps = 60 versus dps = 40): SUBDOM agrees to the printed digits.

Relation to §6.6. The two instruments are complementary: neither supersedes the other. §6.6's mode-shape bar needs FE eigenvectors and, where they exist, its calibration reaches a wider combined evidentiary base (24 candidates at three geometries plus all 24 keys of a $\nu = 0.35$ sector-angle sweep). SUBDOM needs none: it is evaluated from $\mathbf{K}(\Omega)$ alone. On the curated candidate lists above, where both instruments were computed, they agree. The blind test is scored against FE nearest-frequency matching under the inherited block rule, which is a different comparison and is where the remaining discrepancies live. Where an FE model exists, both instruments are available. Where none exists — the situation the flexural screen never faces and the extensional analysis has faced throughout this paper — SUBDOM is, to date, the only two-sided extensional certification available.

6.8 In-plane orthotropic free-free (independent semi-analytical benchmark)

The next three comparisons run against independently published semi-analytical tables: no shared code, mesh, or fitting, and no FE model standing between the solver output and the published number. The extensional (in-plane) spectrum has a first such counterpart: Wang, Shi, Liang and Ahad's modified Fourier–Ritz solution for orthotropic sector plates [10], their Table 4, tabulates the first 8 free-free in-plane frequency parameters at $R_i/R_o = 0.5$, $2\Theta = \pi$ (the same normalized geometry, $r_0/2b = 1.5$, $2\Theta/\pi = 1.0$, as the second adjudication point of §6.5, here with orthotropic material and no physical length scale attached), using the orthotropic material stated in their §3.2: $E_\theta = 70$ GPa, $E_r = 20E_\theta = 1400$ GPa, $\mu_r = 0.3$, $\rho = 7850$ kg/m^3, $G_{r\theta} = (1 - \mu_r\mu_\theta)E_\theta/2 \approx$ 34.84 GPa, from which $\mu_\theta = \mu_r E_\theta/E_r = 0.015$. The same source paper's convergence study prints $E_r = 40E_\theta$; that is a different set and is not used here.

Wang's own nondimensional frequency is

$$\Omega_{\text{Wang}} = \frac{2\omega R_1}{\pi}\sqrt{\frac{\rho(1-\mu_r\mu_\theta)}{E_r}}, \qquad R_1 = R_o,$$

which converts to this project's native raw Ω via

$$K = \frac{\Omega_{\text{Wang}}}{\Omega_{\text{native}}} = \frac{R_o}{b}\sqrt{\frac{c_{66}(1-\mu_r\mu_\theta)}{E_r}} = 0.629609,$$

computed live from the same geometry/material objects entering the search, not a hand-derived constant. All 8 of Wang's tabulated FFFF frequency parameters are matched (Table 9).[13]

[13] An in-run fidelity gate on the isotropic ($E_r = E_\theta$) material reduction at the known FreeFreeIP baseline mode

Table 9: In-plane orthotropic free-free extensional modes vs. Wang, Shi, Liang and Ahad's published semi-analytical solution [10], their Table 4 (Ω_{Wang} convention).

Wang (published)	Solver (Ω_{Wang})	err (%)
1.0311	1.0244	0.649
1.7367	1.7331	0.205
2.0502	2.0435	0.325
3.0618	3.0770	0.498
3.1811	3.1895	0.264
3.4093	3.4229	0.400
4.3023	4.2955	0.159
4.5752	4.5749	0.006

For the in-plane orthotropic case specifically, this closes the "independent exact or published semi-analytical benchmark" gap noted throughout this section for the free-free configuration.

6.9 Isotropic in-plane free-free (independent semi-analytical benchmark)

The isotropic in-plane tables of §6.3 have a second, independent semi-analytical counterpart: Qin, Jin, Chen and Yin's isogeometric solution [8], their Table 2, tabulates the first 6 free-free in-plane frequency parameters at $R_i/R_o = 0.5$, $2\Theta = \pi/2$, the same normalized geometry as the primary benchmark underlying Table 7 itself, $r_0/2b = 1.5$, $2\Theta/\pi = 0.5$, not a second geometry. Qin's other boundary-condition columns are cross-checked in that source against FEM and Ritz; the FFFF 90° column used here is IGA versus Ritz only. Qin's own nondimensional frequency is $\Omega_{\text{Qin}} = \omega R_1\sqrt{\rho/E}$; two alternative forms consistent with the paper's stated general convention were tested and ruled out (0 and 3 of 6 targets within 2%, against 14 candidate matches under the plain form across the recovered spectrum). Converting Qin's 6 published values by the live-computed factor $\Omega_{\text{Qin}}/\Omega_{\text{native}} = 3.896666$ and searching each predicted native location with the production detector recovers 5 distinct genuine roots (log10 $\sigma_{\min} \approx -4.5$ to -5.8, consistent with a real singularity rather than numerical floor) that cover all 6 of Qin's published values (Table 10). For the FFFF 90° column actually used, Qin prints only an isogeometric (Present) column and a Ritz column (his Ref. [25]); there is no FEM column there (FEM appears under CCCC/CFCF). Table 10 lists the IGA values. Qin's Ritz pair at the same geometry is 4.4513 and 4.4627. Converted with the same factor, Table 7 rows 4 and 5 are $\Omega_{\text{Qin}} = 4.4511$ and 4.4623, 0.004% and 0.009% from that Ritz pair and 0.25% apart against Ritz's 0.26%; Qin's IGA pair is 1.04% apart. Those two production roots are the FE pair ($\Omega = 1.142287$ and 1.145158, each FE-matched at 0.006%), and this comparison does not claim whose spacing is physically correct. Against Qin's two IGA targets, a targeted search at each predicted native location, requested n=20 or n=28 with an enlarged $x_{\max}$ that realizes 18 degrees of freedom, recovers a single root covering both ($\Omega_{\text{Qin}} = 4.4529$). The shared $n_{\text{dofs}} = 12$ listing in Table 10 is that IGA comparison; a second isolated dip sits 0.42% away but never surfaces as its own Qin-targeted root, so Qin's IGA pair remains unseparated as two Qin targets.[14] All six published IGA values are accounted for; Qin's close pair is not separated.

Together with §6.8, both the isotropic and orthotropic in-plane free-free spectra now have an independent semi-analytical counterpart. The flexural (out-of-plane) spectrum gains a first one

was confirmed before any orthotropic search ran. Job-level detail: Supplementary Material, §S.4.

[14] Full isolation and MAC detail, including the two isolated frequencies, depths, and their verdicts against distinct, opposite-parity FE targets, confirming two physically distinct shapes despite the shared Qin-targeted root: Supplementary Material, §S.3.4.

Table 10: In-plane isotropic free-free extensional modes vs. Qin, Jin, Chen and Yin's published semi-analytical solution [8], their Table 2 (Ω_{Qin} convention). The solver column is the n=12 shared listing against Qin's IGA (Present) values; the Qin-targeted enlarged-$x_{\max}$ n=20/28 root is $\Omega_{\text{Qin}} = 4.4529$ (18 realized dofs; see text).

Qin (published)	Solver (Ω_{Qin})	err (%)
1.5641	1.5645	0.03
2.6702	2.6791	0.33
2.9792	2.9829	0.12
4.4293*	4.4719	0.96
4.4754*	4.4719	0.08
4.6575	4.6528	0.10

*Both matched by the same n=12 solver root (4.4719), so these two err (%) figures are two readings of one match, not two independent matches. Qin-targeted enlarged-$x_{\max}$ n=20/28 still returns one root ($\Omega_{\text{Qin}} = 4.4529$). Qin's own Ritz pair at this geometry is 4.4513/4.4627, which agrees with Table 7 rows 4–5, not with the IGA pair listed here.

next (§6.10); the primary flexural benchmark tables of §6.2–6.3 themselves remain FE-validated only.

6.10 Isotropic out-of-plane free-free (independent semi-analytical benchmark)

Unlike §6.8 and §6.9, this comparison is flexural (out-of-plane), not extensional: Shi, Shi, Li, Wang and Han's generalized-Fourier-series solution for annular sector plates under elastically restrained edges [15], their Table 3, tabulates six free-free flexural frequency parameters at $R_i/R_o = 0.4$ ($a/b = 0.4$ in their notation), sector angle 90° ($2\Theta/\pi = 0.5$, the primary benchmark's angle at a different radius ratio), isotropic material. It uses the same Ω_{lit} convention as §6.2's orthotropic comparison (Table 5), so unlike §6.8 and §6.9 no paper-specific conversion factor is needed; scale-invariance of the conversion was confirmed numerically before the search ran.

Each of Shi et al.'s six published values was converted to a predicted native Ω and searched with a 13-point, ±15% window, scored by $\log_{10}\sigma_{\min}$ of the same production detector used throughout, then refined by golden-section minimization inside a ±2.5% bracket with a bracket-shift replica as the acceptance test (a minimum that moves when its bracket is nudged is the §5 lock-on failure, not a located root). Four converge cleanly to a stable, isolated minimum, reported as refined roots (Table 11).[15] The remaining two needed additional isolation work: 53.649 failed the bracket-shift test and proved to be a genuine near-degenerate pair 0.59% apart, resolved against its deeper, closer member; 38.428 was a basis-depth near-miss at the production screening basis, not a null result, and cleared the acceptance bar at higher basis size. Both resolve to stable, genuine refined roots reported in Table 11. Full isolation detail, including the second root retained for 53.649 and the basis-convergence trace for 38.428: Supplementary Material, §S.3.5.

All six targets now carry a refined root and a quotable error. It does not close the flexural literature-benchmark gap noted throughout this section, since the primary flexural tables of §6.2–6.3 remain FE-validated only at a different radius ratio; but it is a first data point at a third radius ratio ($R_i/R_o = 0.4$), distinct from both §6.1–6.3's $R_i/R_o = 0.5$ and §6.5's $R_i/R_o = 0.6$.

[15] Acceptance required $\log_{10}\sigma_{\min} \leq -3.5$, error below 2%, and both bracket-shift replicas within 0.1% of the converged root.

Table 11: Out-of-plane isotropic free-free vs. Shi et al. [15], their Table 3 (Ω_{lit}, $R_i/R_o = 0.4$). Err (%) only for bracket-stable refined roots; depth is $\log_{10} \sigma_{\min}$.

Shi et al. (published)	Solver (Ω_{lit})	err (%)	depth	status
15.647	15.6473	0.002	−5.457	refined root
23.576	23.5764	0.002	−4.457	refined root
38.428	38.4295	0.004	−4.706	refined root
53.649	53.6553	0.012	−5.848	refined root*
63.390	63.2956	0.149	−4.534	refined root
70.739	70.7623	0.033	−5.701	refined root

*A second isolated stable root sits at $\Omega_{\text{lit}} = 53.341$ (err 0.574%, depth −3.861); see text.

7 Discussion

Accuracy relative to approximate methods. The FE pair at 459.5/460.6 Hz (Table 7) shows most directly what an exact solution adds over the modified-Fourier–Ritz, isogeometric, and semi-analytical variational frameworks reviewed in §2 [5, 6, 7, 8, 9, 10, 11]: the exact solution splits it into two distinct roots, each matched to 0.006%, using the same 40-digit boundary determinant applied uniformly across the spectrum, with no local mesh or basis refinement targeted at that pair specifically. A basis-truncated method confronting the same splitting would need either an a priori reason to refine locally or a posteriori evidence (e.g. two FE modes suspiciously close in frequency) to know refinement was warranted there at all; the exact method requires neither, because the boundary determinant already resolves whatever structure is physically present. The tabulated values remain subject to the finite retained basis, working precision, and the §5 screen. The extensional tables, which also have independent published semi-analytical checks, are the stronger offering as comparison data; the primary flexural tables remain FE-checked.

Where the exact-reference promise currently holds, and where it does not. §1 motivates an exact solution by a guarantee tied only to retained basis and working precision, with no spatial mesh involved. For flexure that guarantee is delivered in application: the residual screen separates every physical root from every catalogued spurious zero by roughly three orders of magnitude at the production basis (§5), with no overlap across basis sizes 16–44 and the same qualitative split at three further geometries plus the §6.4 sweep (§6.4–§6.5). The numerical cut itself was set by inspection of the FE-labeled physical/spurious populations in the primary tables (§5); once set, applying the screen to a new candidate does not consult an FE model or a published value; FE and the literature are used afterward, to check the flexural tables the screen has already produced. What has been shown to travel is that gap, not a cut derived from the boundary conditions alone. For extension the same guarantee does not yet hold. The pointwise residual is one-sided (§5): it rejects some artifacts with confidence but cannot certify a candidate as real on its own, and the mode-shape bar that closes that gap where an FE eigenvector exists (§6.6) is calibrated against FE eigenvectors at three geometries rather than derived from the boundary conditions alone. The production call at 0.744 does travel: applied unchanged to all 24 keys of the $\nu = 0.35$ sweep it splits every tight candidate with no misclassification wherever a REAL side exists, that is for $2\Theta/\pi \leq 1.0$ (§6.6). What does not travel is the calibration window itself, which narrows under a change of Poisson ratio, radius ratio or sector angle, so the midpoint is a working threshold rather than a derived one. That bar's binding limitation is not portability but provenance: it cannot be evaluated at all without FE eigenvectors.

A second instrument, SUBDOM (§6.7), removes the provenance requirement. Computed from the boundary determinant alone, it is cluster-confirmed at all four of this paper's standard geometries and, on a full blind adjudication set of 122 candidates with no ground-truth label entering the computation, never once accepts a duplicate zero over its FE mode's true owner. Closing the FE-eigenvector-provenance gap (a two-sided extensional criterion that certifies a root from the determinant alone, the way the flexural screen already does) was the principal open problem this paper's earlier drafts left; SUBDOM is offered as that closure. Its limits follow. Basis-size portability at an enlarged basis has one characterised miss (a historical near-degenerate killer whose catalogue Ω remains a local minimum); the near-degenerate block-selection rule it inherits from §6.6's pipeline is now the binding remaining limit at the production basis, having missed two real FE modes that the other parity block at the same Ω matches (one of which recovers at an enlarged basis); the $\nu = 0.35$ geometry's ground truth remains this project's own construction; and working precision is checked on the labelled set at all four geometries. In practical terms, the flexural tables are FE-*checked*, and so, now, can the extensional tables be: FE eigenvectors remain the more evidentiary route where available (§6.6), but the discrimination step itself no longer strictly requires one.

8 Conclusions

A fully-default detector pass is not certified complete (§4): it misses 4 of 26 in-plane modes, later recovered as genuine roots of the same determinant, so the §5 instruments are required for the tables as published. This paper extends the Seok–Tiersten–Scarton exact-wave-function boundary-determinant method from the clamped-free annular sector to the fully free (FFFF) configuration, in both out-of-plane and in-plane motion, as an arbitrary-precision package (Data and code availability).

Six results follow. (i) A from-scratch reimplementation reproduces all 29 tabulated cantilever frequencies (§6.1). (ii) The free-free boundary determinant, with the edge-orientation bookkeeping a naive transplant of the cantilever corner terms gets wrong, decouples into even and odd parity blocks (§3.3); corner-term magnitude and alternation are derived two ways (contour jump; Green's identity), the residual overall-sign label being a convention of the stationarity equation rather than physical content (§3.3). (iii) The residual screen is robust and calibrated for flexure and one-sided for extension; with two companion instruments it catches flexural spurious zeros the papers' double-root criterion misses (§5). (iv) Flexural tables (two-sided, portable screen) agree with FE to 0.64% (isotropic) / 1.08% (orthotropic), with a 6 of 6 literature recovery at a third geometry (§6.2–§6.3, §6.10). Extensional tables (one-sided, geometry-specific) list 26 solver roots covering 27 FE frequencies (one root serves the 1114.9/1115.3 Hz pair; worst unique-partner error 0.052%) and two published semi-analytical checks (§6.8–§6.9). A two-sided mode-shape bar, independent of the residual, separates REAL-like from ARTIFACT-like candidates at three $\nu = 0.30$ geometries (§6.6), resolves eight residual-unresolved candidates at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$ (six ARTIFACT-like, two REAL-like), and still splits every tight FE match on all 24 extensional keys of the $\nu = 0.35$ sweep (threshold 0.744; at $2\Theta/\pi = 1.25$ and 1.50 every tight match is ARTIFACT-like). (v) Every REAL-like flexural candidate has its own independent FE partner at three further geometries — while one in-range FE mode, at (0.6, 1.0), has no REAL-like partner in the other direction — plus a coarser 24-geometry, both-motion nearest-neighbor sweep (§6.4–§6.5). (vi) A second, independent extensional instrument, SUBDOM (§6.7), certifies a root as real or spurious from the boundary determinant alone, requiring no finite-element eigenvector at all: cluster-confirmed at all four of this paper's standard geometries on curated candidate lists (50/50 combined), and on a full blind adjudication set of 122 candidates spanning 77 FE modes, resolving 73 with zero duplicate zeros ever accepted, scored against FE nearest-frequency matching and the inherited block rule at $\gamma = 4$.

Independent ANSYS mode-shape checks confirmed three nearest-frequency pairing defects (FE modes 35, 50 and 53), each with a SUBDOM-real replacement root; two further pairing artifacts (FE modes 16 and 30) have no replacement; two block-rule misses (FE modes 32 and 26) are named in §6.7, of which FE mode 26 recovers only at an enlarged basis.

One flexural FE mode above Table 6's window remains an open assignment between solver roots 2.828013 and 2.838515 (§6.3). The mode-by-mode tables cover only the 2×2 of $\{R_i/R_o = 0.5, 0.6\} \times \{2\Theta = \pi/2, \pi\}$ at $\nu = 0.30$; that standard off this rectangle, and any in-plane residual-screen calibration beyond FF-P1, remain open. The mode-shape bar of §6.6 is not a residual bar, and its $\nu = 0.30$ window is not a universal floor. SUBDOM's own limits — one characterised basis-portability miss and the inherited block-selection rule's two remaining misses (FE modes 32 and, short of an enlarged basis, 26) — are detailed in §6.7 and §7, along with the resolved FE 23 knife-edge and the two pairing artifacts (FE modes 16 and 30) with no replacement found.

The rectangular FFFF extension is not a transplant of the present assembler: the axial solution is closed-form rather than Frobenius, and four free corners replace the sector's two. That case — the four-corner Kirchhoff checkerboard, a persistence screen in place of the two-edge residual taxonomy, and independent flexural and in-plane literature tables — is the subject of a companion paper [18]. The natural sequel of both papers together is general boundary conditions beyond the two classical extremes treated here (simply supported and elastically restrained edges), then functionally graded materials, in-plane orthotropy, piezoelectric constitutive coupling and the inverse use of exact frequencies to identify those constants, multi-input multi-output excitation, and transient shock response.

Data and code availability

The solver, finite-element decks, validated tables, and from-scratch regression suite are released under an MIT license as a public GitHub repository, `https://github.com/McCune1/plate_solver` (tag `v1.0.0`), matching this manuscript; a Zenodo DOI will be added at submission. Tables in §6 were captured or revalidated under `SOLVER_VERSION 2026-07-10.s10`. Results under a different version should be checked against the §4 regression gates first. The free-edge derivation is Supplementary Material, §S.2; conversion constants are §S.3.6. Reproduction scripts and run logs are catalogued in `validation/README.md`; finite-element decks are in `ansys/`.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgments

The computation for this work was performed on The Mill HPC cluster at Missouri University of Science and Technology [19], using up to 32 CPU cores per job. GPU resources were not used.

CRediT authorship contribution statement

Garrek McCune: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Daniel Stutts:** Conceptualization, Supervision.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used generative AI tools to assist with manuscript and code organization, drafting, editing, and formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of this publication.

References

[1] J. Seok, H.F. Tiersten, "Free vibrations of annular sector cantilever plates. Part 1: out-of-plane motion," J. Sound Vib. 271 (2004) 757–772. https://doi.org/10.1016/S0022-460X(03)00414-0.

[2] J. Seok, H.F. Tiersten, "Free vibrations of annular sector cantilever plates. Part 2: in-plane motion," J. Sound Vib. 271 (2004) 773–787. https://doi.org/10.1016/S0022-460X(03)00415-2.

[3] J. Seok, H.F. Tiersten, H.A. Scarton, "Free vibrations of rectangular cantilever plates. Part 1: out-of-plane motion," J. Sound Vib. 271 (2004) 131–146. https://doi.org/10.1016/S0022-460X(03)00365-1.

[4] J. Seok, H.F. Tiersten, H.A. Scarton, "Free vibrations of rectangular cantilever plates. Part 2: in-plane motion," J. Sound Vib. 271 (2004) 147–158. https://doi.org/10.1016/S0022-460X(03)00366-3.

[5] Q. Wang, D. Shi, Q. Liang, X. Shi, "A unified solution for vibration analysis of functionally graded circular, annular and sector plates with general boundary conditions," Compos. Part B: Eng. 88 (2016) 264–294. https://doi.org/10.1016/j.compositesb.2015.10.043.

[6] Q. Wang, D. Shi, Q. Liang, F. Ahad, "An improved Fourier series solution for the dynamic analysis of laminated composite annular, circular, and sector plate with general boundary conditions," J. Compos. Mater. 50 (2016) 4199–4233. https://doi.org/10.1177/0021998316635240.

[7] H. Li, F. Pang, X. Wang, S. Li, "Benchmark Solution for Free Vibration of Moderately Thick Functionally Graded Sandwich Sector Plates on Two-Parameter Elastic Foundation with General Boundary Conditions," Shock Vib. 2017 (2017), Article ID 4018629. https://doi.org/10.1155/2017/4018629.

[8] X. Qin, G. Jin, M. Chen, S. Yin, "Free In-Plane Vibration Analysis of Circular, Annular, and Sector Plates Using Isogeometric Approach," Shock Vib. 2018 (2018), Article ID 4314761. https://doi.org/10.1155/2018/4314761.

[9] X. Shi, D. Shi, Z. Qin, Q. Wang, "In-Plane Vibration Analysis of Annular Plates with Arbitrary Boundary Conditions," Sci. World J. 2014 (2014), Article ID 653836. https://doi.org/10.1155/2014/653836.

[10] Q. Wang, D. Shi, Q. Liang, F. Ahad, "A unified solution for free in-plane vibration of orthotropic circular, annular and sector plates with general boundary conditions," Appl. Math. Model. 40 (2016) 9228–9253. https://doi.org/10.1016/j.apm.2016.06.005.

[11] X. Guan, B. Qin, R. Zhong, Q. Wang, "A unified solution for in-plane vibration analysis of composite laminated sector and annular plate with elastic constraints," J. Low Freq. Noise Vib. Act. Control 40 (2021) 1764–1779. https://doi.org/10.1177/1461348421994419.

[12] W.M. Lee, J.T. Chen, "Analytical study and numerical experiments of true and spurious eigensolutions of free vibration of circular plates using real-part BIEM," Eng. Anal. Bound. Elem. 32 (2008) 368–387. https://doi.org/10.1016/j.enganabound.2007.09.004.

[13] J.T. Chen, J.W. Lee, I.L. Chen, P.S. Kuo, "On the null and nonzero fields for true and spurious eigenvalues of annular and confocal elliptical membranes," Eng. Anal. Bound. Elem. 37 (2013) 42–59. https://doi.org/10.1016/j.enganabound.2012.08.004.

[14] D. Shi, Q. Liang, Q. Wang, X. Teng, "A unified solution for free vibration of orthotropic circular, annular and sector plates with general boundary conditions," J. Vibroengineering 18 (2016) 3138–3152. https://doi.org/10.21595/jve.2016.17004.

[15] D. Shi, X. Shi, W.L. Li, Q. Wang, J. Han, "Vibration Analysis of Annular Sector Plates under Different Boundary Conditions," Shock Vib. 2014 (2014), Article ID 517946. https://doi.org/10.1155/2014/517946.

[16] ANSYS, Inc., *ANSYS Mechanical APDL*, Release 2025 R1, ANSYS, Inc., Canonsburg, PA, USA (2025). Used for finite-element validation only.

[17] R.J. Allemang, "The modal assurance criterion – twenty years of use and abuse," Sound Vib. 37(8) (2003) 14–21.

[18] G. McCune, D. Stutts, "Exact wave-function solution of completely free rectangular plates: four-corner Kirchhoff jump and mode screening," companion paper, submitted simultaneously to this journal (2026).

[19] Missouri University of Science and Technology, "The Mill HPC Cluster," *The Mill* (2024), Article 1. https://scholarsmine.mst.edu/the-mill/1, https://doi.org/10.71674/PH64-N397.